

Review Article

Review on grid-tied modular battery energy storage systems: Configuration classifications, control advances, and performance evaluations



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ABSTRACT

In the past decade, the implementation of battery energy storage systems (BESS) with a modular design has grown significantly, proving to be highly advantageous for large-scale grid-tied applications. However, despite its increasing prevalence, there is a noticeable absence of review papers dedicated to this specific topic. This paper aims to address this gap by presenting a comprehensive overview of modular BESS, covering various aspects such as the classification of modular electrical configurations into four types based on parallel or cascaded connections at the dc or ac side, and providing real-world examples of in-field applications. Furthermore, the paper summarizes the diverse control methods employed in modular BESSs, including power flow control, fault-tolerant control, and battery balancing control. It also presents detailed performance evaluations for different modular BESS configurations, encompassing aspects such as efficiency, cost, battery management, etc. Additionally, this review paper discusses prospective avenues for future research in the field of modular BESSs, offering valuable insights and guidance for their development and application. By providing a comprehensive analysis of modular BESS for practicing battery engineers and aspiring researchers, this paper contributes to the understanding and advancement of this technology, thereby facilitating its integration and utilization in forthcoming applications.

1. Introduction

The grid-tied battery energy storage system (BESS) can serve various applications [1], with the US Department of Energy and the Electric Power Research Institute subdividing the services into four groups (as listed in Table 1) [2]. Service groups I and IV are behind-the-meter applications for end-consumer purposes, while service groups II and III are in-front-of-the-meter applications primarily intended to support the grid.

Due to the electrochemical properties of battery materials, the nominal voltage of the manufactured battery cell is just about 1 V–4 V. Typical nominal capacities could range from several to even a few

thousand ampere-hours. To meet the power and energy requirements of medium-voltage (MV, 3.3 kV and above) ac grid-tied MW/MWh level BESS, a large-scale battery stack is required, as shown in Fig. 1. Battery cells firstly connect in series or parallel to form a battery module (nominal voltage 48 V–100 V, nominal capacity 1 kWh–10 kWh), and then multiple modules connect in series to form a battery rack or cluster (nominal voltage 700 V–1.5 kV), and finally multiple battery racks connect in parallel to form a battery stack with a nominal capacity of 50 kWh–2.5 MWh [3]. However, the large-scale battery stack has voltage and capacity upper limitations.

On the one hand, the number of series-connected cells/modules should not be too large to avoid inconsistency among cells and modules,

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Table 1
Four groups of electric grid energy storage services [2].

Group	I. Bulk Energy Services	II. Ancillary Services	III. Transmission/ Distribution Infrastructure Services	IV. Customer Energy Management Services
Services	Electric Energy Time-shift Electric Supply Capacity	Regulation Reserves Voltage Support Black Start	Upgrade Deferral Congestion Relief Other Related Uses	Power Quality Power Reliability Retail Energy Time-Shift Demand Charge Management

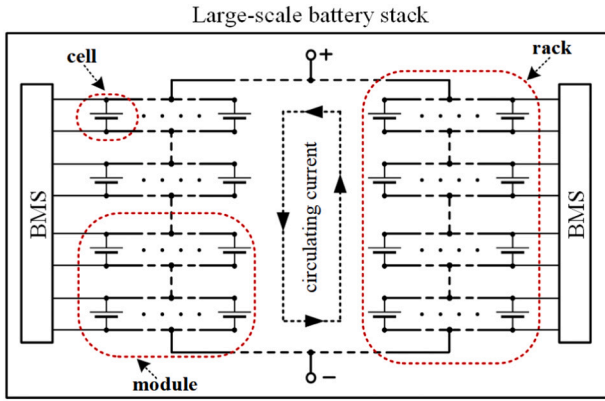


Fig. 1. Configuration of the large-scale battery stack.

which could reduce the burden of the battery management system (BMS) because most commercial BMSs' practical balancing capability is quite limited by using dissipative balancing topologies. Furthermore, the use of too many series-connected cells/modules could lead to poor insulation and safety hazards at connection points, such as arcing or electric shock. To extinguish dc arcing that has no zero-crossing points, expensive MV dc circuit breakers should be used, which will result in a much higher total cost. Moreover, larger series-connection lengths will increase the system's internal resistance due to longer cables and more contactors [4]. Therefore, according to the IEC standard, the maximum voltage of a battery stack is recommended to be below 1.5 kV [5].

On the other hand, the number of parallel-connected racks in a battery stack is also limited. In real applications, the actual open-circuit voltage of each rack is different. Therefore, as shown in Fig. 1, there are circulating currents with a few amperes to several tens of amperes among racks, resulting in cell overcharging/discharging, efficiency reduction, and temperature rise [3].

Over the past decade, modular or reconfigurable configurations [6,7] have become increasingly prevalent and remarkably advantageous in

large-scale grid-tied BESS connected to MV grids, primarily due to the constraints imposed by a single battery stack's limited series and parallel connections. Modular design enables capacity expansion, enhances fault redundancy, and facilitates maintenance, all while ensuring uninterrupted power operation of the BESS.

Typical submodule (SM) structures for the modular BESS are shown in Fig. 2, referring to the dc output and ac output types, respectively. In each SM, very few parallel-connected racks, even a single battery rack or module, are used as a battery pack to decrease the burden of BMS, reduce the circulating current and improve the efficiency. As for the power conversion system (PCS), buck/boost topology [8] and dual-active-bridge (DAB) topology [9] are typically used in the dc-dc converter, while for the dc-ac converter, the low-voltage (LV) two-level (2L) converter and three-level (3 L) neutral-point converter (NPC) are normally used [10]. Then, many SMs will be connected with filters, transformers, and other equipment together to serve as the grid-tied BESS.

To date, several reviews have been published on grid-tied BESS, which are listed in Table 2 [3,8,10–17]. These reviews primarily delve into three main areas: PCS topologies, control methods, and BESS applications. For instance, Wang et al. [10] and Xavier et al. [13] conduct thorough analyses and comparisons of grid-tied PCS topologies, taking into account efficiency, power quality, and the number of components through insightful case studies. Various grid-tied power control methods for BESS are also summarized by other researchers [3,14]. Eroğlu et al. [15] thoroughly examine BESS configurations based on multi-level converters and their accompanying battery balancing control methods. Regarding BESS applications, Hesse et al. [12] offer a comprehensive guideline for selecting the most suitable battery technology, system design, and operational strategies for Li-ion-based BESSs to align with specific requirements. Stecca et al. [16] further summarize the diverse functionalities grid-connected BESSs can provide, including their sizing, location, and control within the distribution network. Lastly, Rouholamini et al. [17] review critical aspects such as battery modeling, BMS architecture, the integration of BESSs into electricity market services, global utility-scale battery storage facilities, and the challenges associated with implementing and managing BESSs.

However, to the best of our knowledge, a review article with comprehensive comparisons is currently lacking, specifically on the large-scale BESS with modular configurations. As mentioned earlier,

Table 2
The classification of existing reviews.

Type	Ref.	Main contributions
PCS topologies	[3,8,10,11,13]–[16]	Summarized different PCS topologies based on various classifications such as conversion stages [3,8,11] or voltage levels [10,13]–[16].
Control methods	[3,14,15]	Reviewed various grid-tied power control [3,14], and balancing control methods [15].
BESS applications	[12,16,17]	Provided the systematic analysis of the technical concepts of battery technologies, system designs, and services.

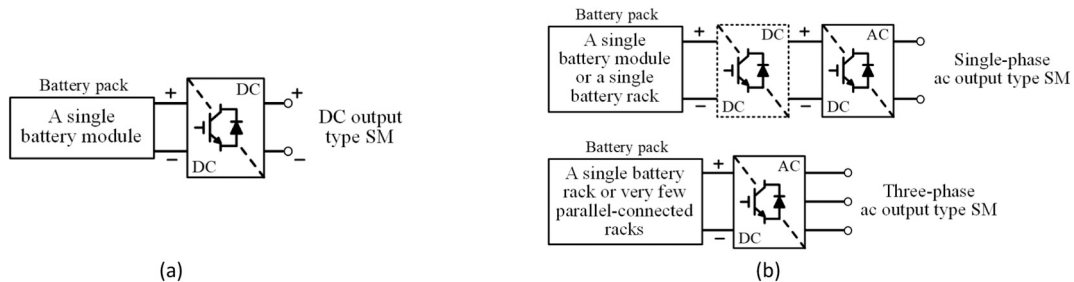


Fig. 2. Two typical SM structures for modular BESSs: (a) dc output type and (b) ac output type.

modular BESSs are more widely implemented in real-world applications and hold significant value for comprehensive summarization. Moreover, for battery professionals intrigued by modular BESSs but lacking sufficient knowledge in grid-tied configurations and control, obtaining a concise overview without getting overwhelmed by the extensive literature available can be challenging. Hence, the primary objective of this paper is to address this review gap by offering a comprehensive analysis of the corresponding configuration classifications and control advances in a condensed and accessible format with real-world in-field applications provided. Additionally, detailed performance evaluations are conducted to provide valuable insights and practical guidance for future applications. This paper is anticipated to be of great interest to a wide audience, particularly practicing engineers and aspiring researchers in the related field.

In Section 2, modular BESS configurations are classified by parallel or cascaded connection at the dc or ac side into four types, and the characteristics of each type are introduced with in-field examples individually. Section 3 comprehensively summarizes the different control methods of different modular BESSs. Section 4 makes performance comparisons for different modular BESS configurations. Finally, the paper concludes with a discussion on future directions in this research field.

2. Configuration classifications of modular BESSs

Connection type among SMs is an important distinction for modular BESSs, especially for in-field applications. Parallel/cascaded connections will influence efficiency, cost, electric isolation, dc grid access, and even control flexibility, consistency, rapidity, and stability. Therefore, modular BESSs will be reviewed due to SM connection types.

As shown in Fig. 3, modular grid-tied BESSs can be divided into four types according to the different combinations of parallel/cascaded SMs at the dc or ac side. Among them, the dc-side-cascaded type is also called a BESS based on the modular dc-dc converter (MDDC), and the ac-side-cascaded type can be mainly divided into two types based on cascaded H-bridge (CHB) and modular multilevel converter (MMC). This section will summarize related research on these four configurations respectively.

2.1. Configurations with parallel SMs

Many industrial applications are using the flexible parallel modular configuration, making it relatively mature and rich in operating experience.

2.1.1. DC-side parallel modular BESS

The corresponding configuration of the dc-side-parallel-connected modular BESS is shown in Fig. 4 [18]. Mattern et al. [19] adopted the buck/boost topology in the dc-dc converter and used this configuration to integrate multiple independent sodium-sulfur battery packs. This configuration has the LV dc bus, which can connect to the local LV dc grid and other distributed dc sources and loads, such as the photovoltaic

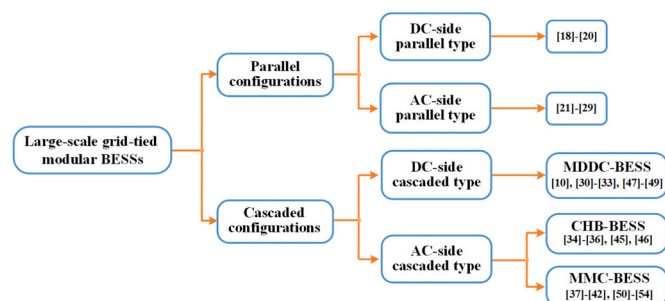


Fig. 3. Four major types of grid-tied modular BESS.

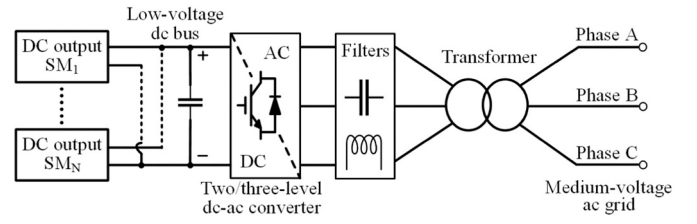


Fig. 4. Configuration of the dc-side-parallel modular BESS.

(PV) panels and electric vehicle (EV) dc chargers.

However, there are also circulating currents among parallel-connected SMs, which reduces the efficiency and may cause a battery pack to be over-charged/discharged, resulting in serious safety problems. Tan et al. [20] used the bidirectional isolated dc-dc topology, DAB, in each SM to eliminate circulating currents among SMs, optimize the control of dc bus voltage and improve the operation efficiency. Nevertheless, the DAB converter will increase the cost significantly compared to the non-isolated dc-dc converter.

2.1.2. AC-side parallel modular BESS

Configuration of the ac-side-parallel-connected modular BESS is shown in Fig. 5 [21]. There are circulating currents among the three-phase ac output type SMs in this configuration, and each SM could be connected to an individual split secondary winding of the multi-winding transformer to eliminate circulating currents [22]. However, the number of secondary windings of a single multi-winding transformer is limited, and therefore many multi-winding transformers are often needed in large-scale BESS, which also increases the cost.

2.1.3. In-field applications of parallel modular configuration

At present, parallel configurations are used in most real-field BESS projects. The “Modular Multi-Megawatt Multi-Technology Medium Voltage - Battery Storage System” (M5BAT) [23] conducted by RWTH Aachen University in Aachen, Germany, in 2016 is introduced here as an example.

The project M5BAT includes planning, building, and operating a 6 MW/7.5 MWh BESS comprising 10 SMs with five different battery technologies, including two different lead-acid technologies and three different Li-ion technologies [24]. The used lead-acid technologies can be divided into lead-acid batteries with liquid electrolyte (OCSM) and sealed lead-acid batteries with gelled electrolyte (OPzV). The installed Li-ion batteries are batteries based on a blend cell chemistry with Li-ion manganese oxide and nickel-cobalt-manganese (LMO/NMC), Li-ion iron phosphate (LFP) and Li-ion titanate oxide (LTO).

Ten battery packs with a dc output voltage between 500 V and 1 kV are installed. The dissipative balancing based on the resistor is adopted in the BMS for each Li-ion battery pack. Each battery pack is connected to a dc-ac converter as an SM and two SMs are connected to one line-frequency multi-winding transformer. The BESS is directly connected to the medium voltage grid after the transformation from 315 V ac to 10 kV ac by five transformers [24,25].

As for commercial applications, since 2017, M5BAT is used for providing the frequency containment reserve (FCR) service with up to 3

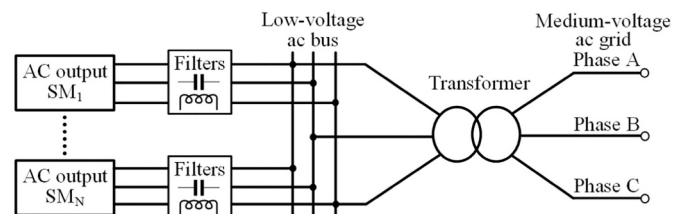


Fig. 5. Configuration of the ac-side-parallel modular BESS.

MW of power. Research field tests were conducted to improve the control algorithms as well as the pilot and testing phase of M5BAT's participation in other markets, such as automatic frequency restoration reserve (aFRR) and energy trading. [26].

Table 3 provides more detailed information on the used components in M5BAT, including each pack's series (s) and parallel (p) interconnection of individual battery cells, power, energy, efficiencies from manufacturers, and updated in-field measurements [27,28]. The analysis of the four-year historical data of M5BAT, operating for the FCR service, has revealed that the whole BESS has an average round-trip efficiency of 72.8 % [27].

More details about M5BAT, including real-time total active/reactive power, grid frequency, battery pack states, and environmental conditions, can be viewed by anyone without authentication from its website [29].

2.2. Configurations with cascaded SMs

With the application of cascaded converters during the past decade, grid-tied modular BESSs with cascaded configurations have attracted extensive research and applications.

2.2.1. DC-side-cascaded modular BESS

Inspired by the grid-tied PV system using cascaded modular dc-dc converters [30], the configuration of the MDDC-BESS is shown in Fig. 6(a). Mukherjee et al. [31] used MDDC with the half-bridge converter in each SM for the second-life power battery utilization. This configuration was also applied to the battery and super-capacitor hybrid energy storage system (HESS) [32], and the topology of using a 3L converter in the dc-dc stage was also explored [33].

The line-frequency transformer could also be eliminated in this kind of configuration, as shown in Fig. 6(b). Wang et al. [10] reviewed this transformer-less configuration using the MV 2L/3L dc-ac converter based on the series connection of semiconductors.

2.2.2. AC-side-cascaded modular BESS

This configuration adopts cascaded single-phase ac output SMs to achieve dc-ac conversion. Two typical configurations are shown in Fig. 7, where Fig. 7(a) shows the star-connected CHB-BESS while Fig. 7(b) shows the MMC-BESS. There is another delta configuration for the CHB converter, but the star configuration is preferable in BESS applications to minimize the number of cascaded SMs [34]. Therefore, CHB in this paper refers to the star-connected configuration.

In 2007, Maharjan et al. [35] first proposed to apply CHB into the super-capacitor energy storage system to realize direct integration into the medium voltage ac grid without the line-frequency transformer. Subsequently, they applied this configuration to the NiMH BESS in 2009 [36]. Trintis et al. [37,38] first combined MMC with BESS in 2011. Compared with CHB, MMC has the common dc-link and can work as the

interfacing converter to integrate large-scale energy storage batteries, ac and dc grids [39,40].

As for the dc-ac stage in each SM, topology variations have already been summarized in detail in existing reviews [8,15]. In most cases, CHB-BESS adopts the full-bridge circuit while MMC-BESS adopts the half-bridge circuit, which are all single-phase dc-ac converter topologies.

As shown in Fig. 2(b), single-phase ac output SMs can be divided into two types according to whether they contain the bidirectional dc-dc stage. Soong et al. [41,42] compared the advantages and disadvantages of these two SM types for the ac-side-cascaded modular BESS in detail. Specifically, the single-phase dc-ac stage will induce low-frequency current ripples flowing through batteries, influencing the normal functions of BMS. Such current ripples can be suppressed by controlling the dc-dc converter. Also, the number of cascaded SMs could be further reduced with the dc-dc converter. However, more power switches and passive devices are needed for the dc-dc converter, increasing the total cost and size.

In addition, researchers used CHB and MMC to integrate different kinds of energy components with batteries together, such as electrical double-layer capacitors (EDLCs) [43] and PV panels [44].

2.2.3. In-field applications of cascaded modular configuration

CHB-BESS has already been rolling out among cascaded modular configuration candidates in engineering applications. In these applications, the full-bridge circuit was used for the dc-ac stage and there was no dc-dc stage in each SM.

Kawakami et al. [45] built the first 500 kW/238 kWh lithium-ion CHB-BESS in 2014 that could connect to the MV ac grid. In particular, 27.6 V/20 Ah LTO battery cells were 2-in-parallel and 12-in-series (2p12s) connected to get the 331.2 V/40 Ah battery pack for each SM. Six SMs were needed for each phase in total. However, the output voltage of this CHB converter was only 1.5 kV, and it still used a line-frequency transformer to connect to the 6.6 kV grid.

In September 2014, Cai et al. [46] implemented the world's first demonstration application of a 2 MW/2MWh transformer-less 10 kV grid-tied CHB-BESS at the Baoqing energy storage power station in Shenzhen, China. Detailed parameters of this in-field application are shown in Table 4. This CHB-BESS is mainly used for electric energy time-shift for the local grid. The PCS's efficiency was measured to be >98 % and the full-cycle efficiency of the whole BESS was higher than 90 % when operating around the nominal operating point.

MDDC- and MMC-BESS remain mainly in laboratory prototypes and have not yet been applied in practical engineering applications. For MDDC-BESS, in the research project "Highly Efficient and Reliable Modular Battery Energy Storage Systems" conducted by RWTH Aachen University [47], the dc-ac converter adopting medium voltage components and 3 L active NPC topology was proposed to connect the 4.16 kV or 6.6 kV ac grid directly [48]. The DAB topology was used for the

Table 3
Technical information on the different components in M5BAT [27,28].

Components		Technology		Power (kW) ¹	Capacity (Ah) ²	Efficiency (%) ³	
						From manufacturers	In-field measurements
Battery Packs in 10 SMs	1–2	OCSM	300s1p	617	1776	75.0	86.0–88.0
	3	OPzV	308s2p	617	1368	85.0	81.3
	4	OPzV	306s1p	512	1209	85.0	81.7
	5–8	LMO/NMC	192s16p	617	1088	97.0	98.0–98.5
	9	LFP	240s10p	617	933	95.0	94.2
	10	LTO	312s32p	617	320	> 96.0	96.0
PCS (DC-AC Converter)		SMA Sunny Central 630CP XT		630	/	98.6	94.1
Transformer		Schneider HSGX 1260F		1260	/	98.0	94.6

¹ The power of battery packs is measured from the ac-side, including the average losses of the dc-ac converter and the transformer.

² The capacity of battery packs is specified for a C-rate of 1/3.

³ The in-field round-trip efficiency is measured for battery packs, while the in-field power efficiency is measured for the dc-ac converter and the transformer using the four-year historical data operating for the FCR service.

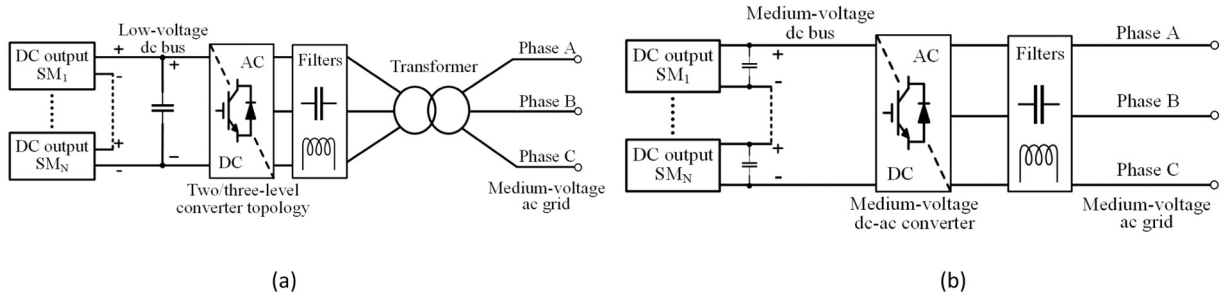


Fig. 6. Configuration of the dc-side-cascaded modular BESS (MDDC-BESS) (a) with and (b) without the line-frequency transformer.

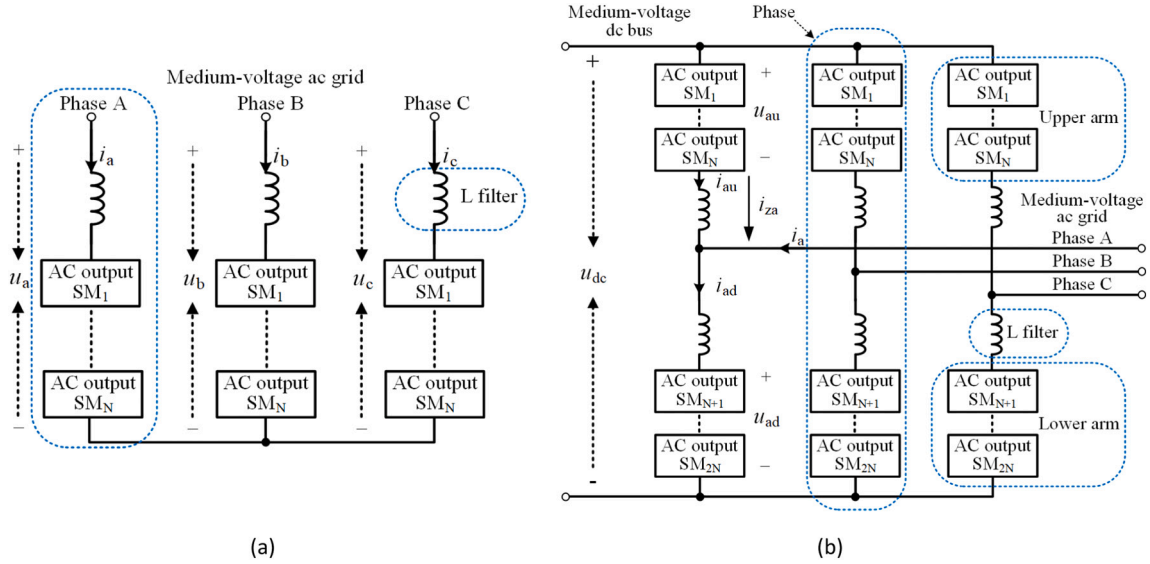


Fig. 7. Configuration of the ac-side-cascaded-connected modular BESSs: (a) CHB-BESS (b) MMC-BESS.

Table 4

Technical information on CHB-BESS at the Baoqing energy storage power station [46].

Battery cell parameters		The battery pack of each SM				Total SM number	Efficiency (%) ¹		
Chemistry	Nominal parameters	Cell connection	Voltage (V)	Capacity (Ah)	Power (kW)		Battery	PCS	BESS
LTO	2.3 V/50 Ah	294 s	672	50	33.6	60	Not provided	98	90

¹ The in-field power efficiency is measured for PCS when operating from 0.6 to 1 p.u. The in-field round-trip efficiency is measured for battery packs using the application-independent full-cycle.

cascaded dc-dc converters, and the total dc output voltage was adaptable to different medium dc grid voltage levels [49]. For MMC-BESS, Hillers et al. [50–54] conducted an in-depth system study and compared it with CHB-BESS. They built a 250 kW prototype with 90 SMs in the laboratory, which can realize the direct connection to the 9 kV grid. Features of MDDC- and MMC-BESS about coupling ac/dc conversion stations and energy storage have attracted great attention from the electricity transmission system operators.

3. Control advances of modular BESSs

This section reviews the three most important topics about modular BESS operation control, including power flow control, fault-tolerant control, and battery balancing control. Power flow control manipulates the active and reactive power of each SM during normal operating conditions. Fault-tolerant control ensures the BESS continues to work normally when some components inside the SM fail. Battery balancing control deals with battery state imbalances among SMs to maximize the available capacity and lifetime.

3.1. Power flow control

Power flow control is the basis for the normal operation of the modular BESS. Fig. 8 shows the main reviewed contents of the power flow control of modular BESSs.

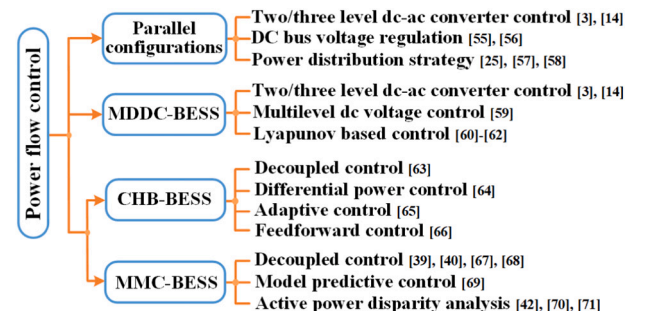


Fig. 8. Main contents for the power flow control of modular BESSs.

For parallel configurations and MDDC-BESS, the commonly used LV 2L/3L dc-ac converter is normally responsible for the dc-ac power flow control. The corresponding control methods are comparatively commercially mature and have been well summarized in [3,14], including proportional-integral (PI) control, proportional-resonant (PR) control, predictive control, direct power control, repetitive control, grid-connected/islanding control. Therefore, power flow control research reviewed in this section for these two configurations focuses on the dc-side collaborative control and the power distribution strategy among SMs to get extra benefits.

For the ac-side cascaded type configuration, CHB or MMC is adopted as the dc-ac converter and each SM cannot do power flow control independently. Thus, there is a coupling relationship among them and most power flow control methods are more complex and will be reviewed in this section.

3.1.1. Parallel configurations

For the BESS based on the dc-side-parallel modular BESS, Rehman et al. [55] used differential power sharing to realize dc bus voltage regulation and output voltage sharing among SMs under bidirectional power flow. Gao et al. [56] also proposed a variable dc bus voltage control method to obtain the lowest power loss.

For the power distribution strategy, Choi et al. [57] proposed the multiplicative-increase-additive-decrease principle in the HESS based on the dc-side-parallel configuration so that it can achieve minimum magnitude/fluctuation of battery current and energy loss. Tummuru et al. [58] proposed a new power management algorithm that could set the system operation mode based on the changes in renewable energy sources and the load connected to the LV dc bus. Based on the MSBAT, Ref. [25] presented a rule-based power distribution algorithm to provide the German power grid frequency containment reserve. In this way, the state of charge, energy throughput, and power load of the batteries can be controlled to extend their lifetime.

3.1.2. DC-side-cascaded modular BESS (MDDC-BESS)

In contrast to the traditional constant dc bus voltage control, Zheng et al. [59] used MDDC with the half-bridge circuit to achieve variable multilevel dc voltage output, and the subsequent full-bridge dc-ac converter can only commutate the current at a low switching frequency, reducing the switching losses in the dc-ac stage.

Mukherjee et al. [60] found that the variation of battery states in different SMs could give rise to control stability problems when using the conventional cascaded PI control-loop approach with fixed controller parameters. Therefore, they further explored the corresponding control solution based on Lyapunov functions [61]. In addition, they presented an analysis and comparative study of all the working modes of MDDC-BESS along with switching performances in detail to help select suitable converter modes [62].

3.1.3. AC-Side-Cascaded Modular BESS

a) CHB-BESS

For CHB-BESS, Maharjan et al. first proposed a decoupled control method, which greatly improves battery energy utilization [63]. Liu et al. proposed to use the differential power control method [64] and adaptive control method [65] in CHB-BESS to integrate the retired electric vehicle batteries.

Besides, Jiang et al. [66] proposed a feedforward control based on the space-vector-modulation (SVM) to distribute the active power between the battery and the EDLC.

b) MMC-BESS

In MMC-BESS, both ac and dc grids can exchange bidirectional power with the BESS, while the power transfer between ac and dc grids

can be decoupled [39,67]. Fig. 9 shows the control block diagram of ac and dc power flow control based on PR and PI controllers. In Fig. 9(a), firstly, the d-axis component (u_d) of the three-phase output voltages (u_{x0}) and d/q-axis components (i_d and i_q) of the three-phase output currents (i_x) are derived from the Clarke/Park transformation and phase-locked loop (PLL). Then, the current references i_d^* and i_q^* are calculated from the ac active/reactive power flow references P_{ac}^* and Q_{ac}^* . Errors are transformed back to the three-phase frame and a PR controller (PR_{h1}), according to the fundamental frequency, is used to control them. In Fig. 9(b), the dc circulating current i_{zx-dc} could be used to do the dc power flow control; it can be obtained by applying a low-pass filter (LPF) to the circulating current i_{zx} . The controlling reference i_{zx-dc}^* could be calculated by the dc power flow reference P_{dc}^* and the dc grid voltage u_{dc} . The PI controller is used to track i_{zx-dc}^* . In addition, Zhang et al. proposed a decoupled power control method for MMC-HESS with batteries and super-capacitors [40,68].

Li et al. also used model predictive control in MMC-BESS, which can deal with dc/ac power transferring without decoupling [69]. As for the control limitation, Liang et al. [70] presented a general analysis of the active power difference among SMs, which incorporates the inherent operational constraints of MMC. The proposed analysis can help facilitate the design of MMC-BESS for applications that require significant inter-SM active power disparity. Similar power difference analysis has also been carried out in other research [42,71].

3.2. Fault-tolerant control

The ability to perform fault-tolerant control by setting redundant SMs is a major advantage of the modular configuration, and it is especially important for cascaded modular configurations where charging/discharging current coupling relationships exist among SMs. According to the redundant SM type [72], fault-tolerant control can be divided into two kinds: the cold reserved SM method and the hot reserved SM method, as demonstrated in Fig. 10.

3.2.1. Cold reserved SM method

The cold reserved SM method is relatively simple. Using this method, when the system operates normally, the redundant SM is bypassed (using its own PCS circuit or additional bypassing switch). When an SM fails, the redundant SM is triggered to replace the failed SM [73]. Bi et al. [74] proposed a fast diagnostic method for SM open-circuit faults of super-capacitor energy storage systems based on the MDDC, and also a redundant control strategy based on cold-reserved SMs. However, the states of the battery pack in the cold reserved SM may differ from the battery states of operating SMs, which will cause a transient switching process and affect the output power quality. Moreover, cold reserved SMs are not involved in regular operation, which is uneconomical, with low semiconductor and battery utilization ratios [75].

3.2.2. Hot reserved SM method

In the hot reserved SM method, redundant SMs are put into a working state during normal operation. When faulty SMs are cut off, corresponding control methods should be taken to keep the system's normal operation. As a result, this method has a high SM utilization ratio. It also has good state consistency among SMs and better transient switching performance. Furthermore, it can improve BESS's maximum output voltage and power and enhance the system's resilience to grid overvoltage risk. The hot reserved SM method can be divided into two types depending on how to deal with faulty SMs.

a) Bypassing extra normal SMs

The first type of the hot reserved SM method is to bypass the same number of normal SMs in other arms/phases when bypassing faulty SMs in one arm/phase. This method can ensure that three-phase outputs are

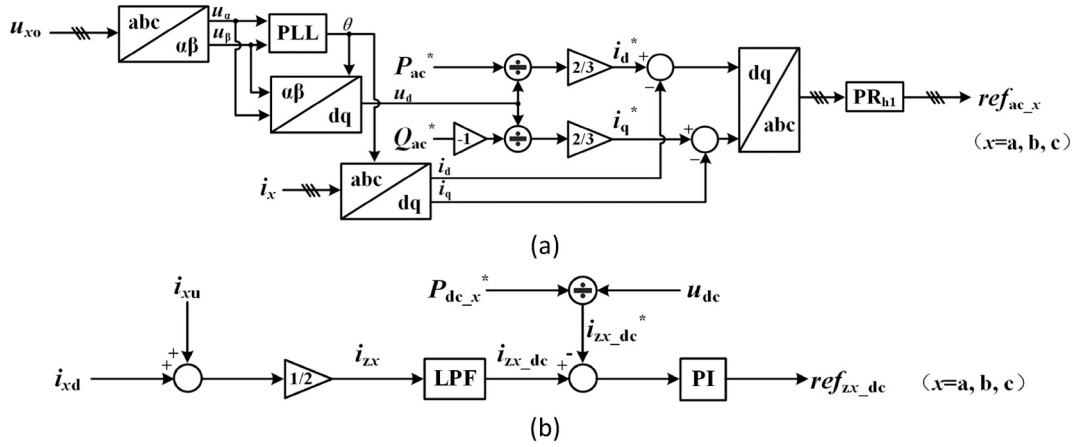


Fig. 9. Decoupled power flow control of MMC-BESS at (a) the ac-side and (b) the dc-side.

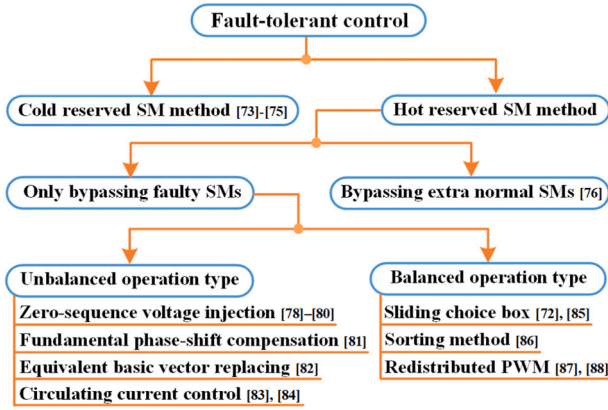


Fig. 10. Main contents for the fault-tolerant control of modular BESSs.

balanced, but it may affect the output power quality and reduce BESS's capacity and reliability [76].

b) Only bypassing faulty SMs

Bypassing faulty SMs without changing other arms/phases is also possible. By doing so, higher redundancy is achieved.

(1) Unbalanced operation type

In principle, only bypassing faulty SMs will lead to the unbalanced operation of the whole system and affect the output power quality directly [77]. The zero-sequence voltage injection method is commonly used to reduce the influence of faulty SMs, which can change the phase voltage without affecting the output line-to-line voltages, line currents, or output power [78–80]. Alenejad et al. [81] proposed a fault-tolerant strategy by implementing the fundamental phase-shift compensation method to restore the operation of the CHB-BESS with one or more faulty switches to the pre-fault conditions. For BESS using the SVM, the equivalent basic vector is used to replace the vector affected by the faulty SMs, thus reducing the output voltage range [82]. For MMC-BESS, Ref. [83, 84] proposed circulating current control methods to solve the problem of unbalanced power distribution among arms when only faulty SMs are bypassed.

(2) Balanced operation type

It is even possible to avoid unbalanced operation when only

bypassing the faulty SM by using the following methods.

Li et al. [72,85] used the rotating sliding choice box to insert the minimum number of SMs required per arm during normal operation. All SMs take turns working in the operating mode and the standby mode, which equalizes the power load of each SM on a relatively large time scale. The failed SM is removed from the selecting option of the sliding box immediately, which guarantees a nearly seamless transition and a very short recovery time.

However, when using the hot reserved SM method for modular BESSs adopting carrier-based pulse width modulation (PWM), once an SM fails and is bypassed, the remaining SMs need to adjust the carrier phase-shifting angle or carrier phase-disposition amplitude, which will increase the system complexity, calculation and communication burden, and the delay for modulation and control.

To overcome the above drawbacks, Wang and Tang [86] proposed a new fault-tolerant method by combining the traditional carrier phase-shifting modulation and the sorting method, eliminating the one-to-one correspondence between carrier and SM. Ref. [87, 88] restored the BESS normal ac output by redistributing the PWM pulses of the faulty SM to other normal SMs (R-PWM method) with the basic working principle explained in Fig. 11. This method can ride through the SM fault by just employing simple logic operations without adjusting the carriers' angles.

3.3. Battery balancing control

Each battery pack in the parallel modular configuration can be controlled independently with high flexibility. The only critical balancing-related aspect of the parallel configuration is the imbalance among series-connected battery cells within the pack, which should be compensated by the internal BMS and has also been reviewed comprehensively. For the cascaded modular configuration, battery packs are coupled and cannot operate independently. In other words, in addition

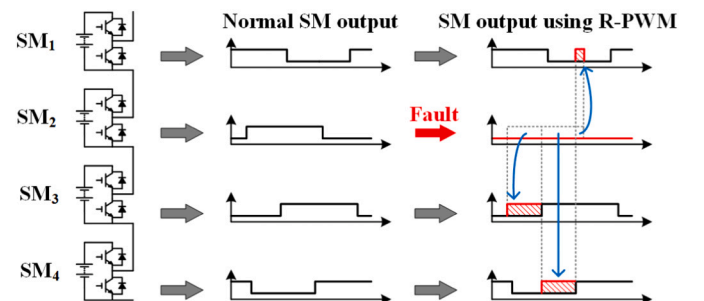


Fig. 11. Basic working principle of R-PWM method [87].

to the internal battery imbalances within each pack, the imbalances among packs should also be taken into account. Therefore, the cascaded modular BESS needs unified hierarchical balancing control.

Fig. 12 illustrates the various causes of imbalances among batteries that can occur in BESS [89]. Two key root causes of internal imbalances are unavoidable differences in battery production and varying operational conditions, such as temperature fluctuations. These factors result in differing initial conditions, including capacity and internal resistance, as well as varying aging and discharging rates, which ultimately lead to deviations in state-of-charge (SOC) and state-of-health (SOH). SOC represents the remaining capacity during operation, and SOC imbalances among cells can result in decreased available pack capacity. On the other hand, SOH represents the remaining life of the battery, and SOH imbalances among cells can lead to a reduction in the total lifetime of the BESS.

Regardless of balancing aims, battery balancing control methods can be divided into inter-SM, inter-phase/arm, and multilayer balancing control. Fig. 13 shows the categorization of different battery balancing control methods.

3.3.1. Inter-SM balancing

Inter-SM balancing is also called intra-phase balancing for CHB-BESS or intra-arm balancing for MMC-BESS. It can be divided into three categories: the carrier rotation method, the sort-and-select method, and the modulation reference adjustment method.

a) Carrier rotation method

The carrier rotation method realizes battery state balancing for the cascaded modular BESS using carrier phase disposition (PD) PWM. In particular, under normal conditions, this method interchanges carrier signals with switching frequency [90] or line frequency [91] among SMs so that equal power sharing among SMs is achieved. Under the battery state unbalanced condition, such as SOC imbalances, the position of each SM's carrier wave is determined by its SOC [59]. During discharging, SMs with higher SOC are placed at the bottom layer of the carrier wave, while SMs with lower SOC are redirected to the top layer. SMs at the bottom layer will discharge more and SMs at the top layer will be used less. Reversely, during charging, SMs with higher SOC will be used less and the battery pack with lower SOC will be charged more.

b) Sort-and-select method

In the sort-and-select method [92,93], all SMs' specific states, such as SOC, are sorted first. Then, the battery pack with the highest SOC will be the first chosen to be discharged and the last chosen to be charged. The application range of this method is wider and more flexible for both carrier-based and nearest-level modulation. However, the calculation of the sorting sequence will consume much more time when the SM number is large.

For MDDC-BESS, Ma et al. [94] proposed a sorting selection balancing control scheme to realize battery pack hot-swapping and SOC

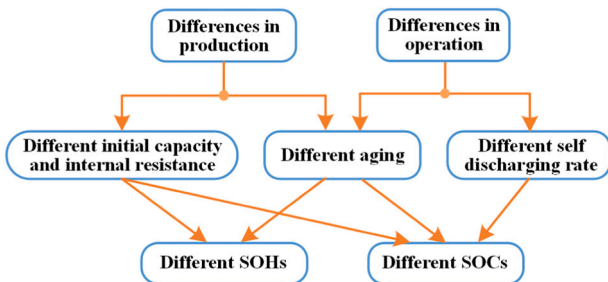


Fig. 12. Causes for the imbalances among batteries [89].

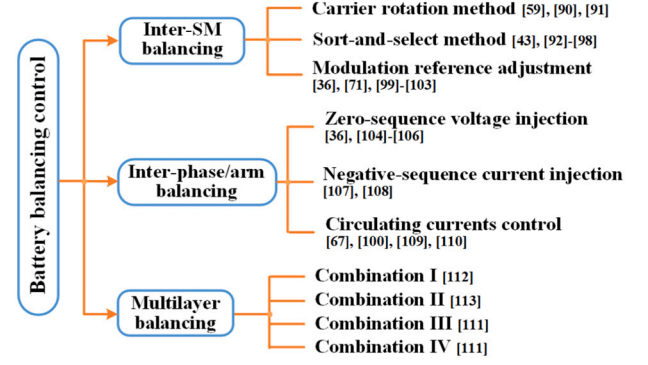


Fig. 13. Main contents for the battery balancing control of modular BESSs.

balancing at the same time.

For CHB- and MMC-BESS, Ref. [95] determined which SMs should be charged/discharged according to the current direction and the output voltage level after sorting SMs' battery pack voltages. This method can ensure that SMs are balanced when the converter switching loss is low. In [96], one SM was bypassed from the system and its pseudo-open-circuit voltage (POCV) was measured every 3 s. After measuring the POCVs of all SMs, they were sorted and the sorting sequence was fed to the nearest-voltage-matching algorithm to do the balancing. Jiang et al. [43,97] used sorting and selection to achieve the SOC balancing by adjusting phase angles of the voltage reference's high-frequency part in each SM. Liang et al. [98] established the sort-and-select method to regulate the SM active power, considering active power constraints.

c) Modulation reference adjustment method

Each SM's modulation reference signal could also be modified directly to control the active power transfer among SMs [36,99]. This method is relatively straightforward. However, directly adjusting each SM's modulation reference will unavoidably worsen the output quality, especially when the batteries' states are highly different [100].

Bi et al. [101] proposed an SOC active balancing control strategy adopting such kind of method for the super-capacitor energy storage system based on MDDC. Chowdhury et al. [102] used the PI controller to track the modified charging/discharging current reference for the bidirectional dc-dc converter in each SM of MDDC to realize SOH balancing. Mukherjee et al. [103] modified each SM's current reference according to the SOC derivation method so that each SM's charging/discharging trajectory finishes at the same time. For MMC-BESS, Wang et al. [71] optimized the ratio of ac and dc modulation indexes to enhance the tolerance of unbalanced power among SMs within each arm, which greatly improved the SOC balancing rate.

3.3.2. Inter-phase/arm balancing

For CHB- and MMC-BESS, there is also inter-phase/arm balancing aiming to make the average SOC/SOH of all SMs in different phases/arms equalized.

The inter-phase/arm battery balancing can be mainly divided into three categories: zero-sequence voltage injection, negative-sequence current injection, and circulating currents control.

a) Zero-sequence voltage injection

This method calculates the corresponding zero-sequence voltage according to the amplitude of phase-to-phase balancing power and then superimposes it on the voltage reference.

Maharjan et al. implemented this method firstly on the CHB-BESS with star configuration [36]. Vasiladiotis and Rufer [104] comprehensively analyzed this method for balancing control in CHB-BESS, considering three-phase unbalanced grid faults. Ref. [105, 106] also

used this method for inter-phase balancing in MMC-BESS under unbalanced grid faults.

b) Negative-sequence current injection

The negative-sequence current injection method will inject a small amount of negative-sequence current into the reference current to adjust the three-phase power distribution [107]. This method does not affect the BESS output power but deteriorates the power quality.

Hatano et al. [108] combined both zero-sequence voltage and negative-sequence current injection methods to deal with inter-phase balancing under unbalanced conditions due to grid faults, which can reduce the dc voltage fluctuation in each SM.

c) Circulating current control

To summarize, the above two kinds of inter-phase balancing control are realized by adding deviation power on the input/output power reference. Therefore, the balancing must be completed during operation and cannot be done in standby mode. In contrast, MMC-BESS can do inter-phase/arm balancing by using dc/fundamental circulating currents even in a standby mode.

Conventionally, the circulating currents of MMC should be limited to improve efficiency and reduce the dc voltage fluctuation. Gao et al. [100] employed circulating currents to do the inter-phase and inter-arm SOC balancing for MMC-BESS. For the inter-phase SOC balancing, the dc component of the circulating current (i_{zx_dc}) is used to distribute power among three phases, as shown in Fig. 14(a). According to the error between the average SOC of all SMs in phase x (SOC_x) and the average SOC of all SMs in the whole BESS (SOC_{ave}), the dc circulating current reference $i_{zx_dc_soc}^*$ for the inter-phase SOC balancing is obtained by the proportional (P) controller. After adding the dc circulating current reference $i_{zx_dc_soc}^*$ from the normal dc power control, the total reference $i_{zx_dc_tot}^*$ is tracked using the PI controller. For the inter-arm SOC balancing, the fundamental-frequency circulating current (i_{zx_hl}) is used to distribute power between the upper and lower arms, as shown in Fig. 14(b). The value of the fundamental frequency circulating current reference $i_{zx_hl_soc}^*$ is obtained by the P controller according to the error between the average SOC of all SMs in the upper arm (SOC_{xu}) and the lower arm (SOC_{xd}). The frequency and phase of $i_{zx_hl_soc}^*$ are obtained by the PLL to synchronize with phase output voltage. Then, PR_{hl} is used to track $i_{zx_hl_soc}^*$.

Li et al. [109,110] first implemented this method to do the SOH balancing by redistributing power demand among phases and arms. In addition, for the MMC-BESS, which interconnects ac and dc grids, Chen et al. [67] systematically summarized SOC balancing control methods among phases, arms, and SMs in normal operating modes, ac fault modes, and dc fault modes, respectively.

3.3.3. Multi-layer balancing control

All methods mentioned above are inter-pack balancing control methods for modular BESSs. Actually, the intra-pack balancing can also be conducted by using the internal balancing circuit of BMS. Therefore,

there are four different battery balancing control strategies in total. They are inter-pack SOC balancing, inter-pack SOH balancing, intra-pack SOC balancing, and intra-pack SOH balancing. Ref. [111] summarized the multi-layer battery balancing control for modular BESSs shown in Table 5, including four strategy combinations. For four different BESS service groups listed in Table 1, the intra-pack and inter-pack balancing for modular BESSs can adjust accordingly to achieve different goals, which means selecting the most effective balancing strategy combination is quite valuable in practice.

Specifically, strategy combination I has been studied for CHB-BESS in [112]. Intra-pack SOC balancing is realized by using two kinds of additional equalization circuits, which will improve the effects of the inter-pack SOC balancing. For strategy combination II, Ref. [113] first combined intra-pack SOH balancing and inter-pack SOH balancing in MMC. MMC coordinates the pack and cell SOH equalization schemes, where the charging/discharging power per SM was properly adjusted according to the extent of SOH deviation inside the pack and the extent of SOH deviation among packs. Finally, Ref. [111] first proposed strategy combinations III and IV with effects analysis and experimental verifications.

4. Performance evaluations of different configurations

Comparisons made in this Section are for the MV (3.3 kV and above) ac grid-tied MW/MWh level BESS. Performance comparisons of four grid-tied modular configurations are summarized in Table 6.

4.1. Efficiency

The parallel modular configuration requires line-frequency transformers for medium voltage ac-grid connection. Moreover, the output power quality of the commonly used LV 2L/3L dc-ac converters in this configuration is poor, and large-capacity filtering devices are required to meet the grid-tied requirements [10]. In addition, compared to the cascaded configuration, the voltage ratio of the dc-dc converter per SM in parallel modular configuration is larger [30]. All of the above issues could influence the system's efficiency.

MDDC-BESS also requires large-capacity filtering devices and most of them still need the line-frequency transformer to connect to the

Table 5

Multi-layer balancing control for cascaded modular BESSs [111].

Control strategy combination	Intra-pack balancing	Inter-pack balancing	Suitable energy storage service group
Combination I	SOC balancing	SOC balancing	I. Bulk Energy Services
Combination II	SOH balancing	SOH balancing	II. Ancillary Services
Combination III	SOC balancing	SOH balancing	III. Transmission/ Distribution Infrastructure Services
Combination IV	SOH balancing	SOC balancing	IV. Customer energy management services

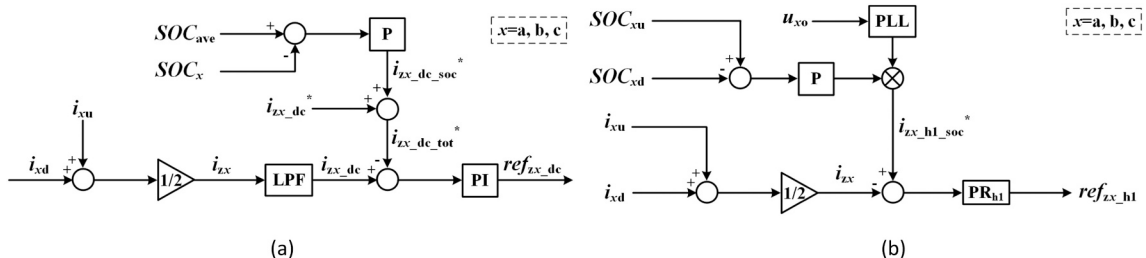


Fig. 14. Control block diagram for the (a) inter-phase and (b) inter-arm SOC balancing in MMC-BESS.

Table 6

Performance comparison of grid-tied configurations of different modular BESSs.

Configurations	Efficiency	Cost	Battery current ripples	Electric isolation	DC grid access versatility	Power flow control	Battery management
DC-side parallel modular BESS AC-side parallel modular BESS	Low	Medium	Switching frequency ripples	Good	LV dc grid None	Flexibility: high; Consistency: low; Rapidly: low; Stability: low.	BMS burden: high; Battery balancing control: independent and flexible.
DC-side cascaded modular BESS (MDDC-BESS)	Medium			Varying	LV/MV dc grid	Flexibility: low; Consistency: high; Rapidly: high; Stability: high.	BMS burden: medium; Battery balancing control: unified and complex control.
AC-side cascaded modular BESS	CHB-BESS	High	Low	Switching frequency and double line frequency ripples	None		Battery balancing control: Strongest ability to balance batteries.
	MMC-BESS	Highest	High	Switching frequency, line and double line frequency ripples	MV dc grid		

medium voltage ac grid. Furthermore, the mandatory dc-dc stage in each SM also decreases the overall efficiency. Fortunately, the voltage ratio of each dc-dc converter is relatively small, which could improve the overall efficiency to a certain extent.

CHB- and MMC-BESS can achieve medium voltage ac grid connection without a line-frequency transformer or large-capacity filtering devices. This configuration can also reduce the working current under the same output power due to its high output voltage. Therefore, it is expected to achieve high efficiency.

Wang et al. [10] theoretically calculated and compared the efficiency of PCSs in different configurations under different output power conditions. Soong and Lehn [41] also compared the efficiency of MMC-BESS and CHB-BESS. These two independent studies have proven that the efficiencies of AC-side cascaded modular BESS are significantly higher than other grid-tied configurations. Furthermore, under heavy load, the efficiency of MMC-BESS is even higher than that of CHB-BESS.

Table 7 compares the reported efficiencies of two reviewed in-field BESS applications with parallel and cascaded modular structures. The reported round-trip efficiency of M5BAT (72.8 %) is much lower than that of Baoqing CHB-BESS (90 %). The large efficiency difference between these two BESSs results from different battery types, commercial applications, efficiency measurement methods, and BESS configurations.

Firstly, as shown in Table 3, more than half of the battery capacity in M5BAT comes from lead-acid batteries, whose efficiencies are much lower than the LTO batteries used in Baoqing CHB-BESS. Next, the efficiency of M5BAT is calculated from the in-field four-year historical data, in which M5BAT suffers from the less efficient operation of PCSs with an average utilization rate of 15.8 % and transformers with an average utilization rate of 9.75 % due to the low power requirements in FCR service [27]. For the Baoqing CHB-BESS, the application-independent full-cycle efficiency around nominal operating conditions was measured for the whole BESS [46]. This also reminds engineers that the modular design should be further utilized to avoid the low power/energy operation point of components in the BESS. Lastly, the combined in-field efficiency of the PCS and the transformer in M5BAT is only 89 %. Even if using the power efficiency from manufacturers in Table 3, the combined efficiency of the PCS and the transformer in M5BAT is only 96.6 %, still lower than that of the transformerless CHB, which is >98.3 % when outputting nominal power [46].

Table 7

Comparisons of two reviewed in-field BESS applications.

Configurations	Projects and Ref.	Total power and energy	Battery type	Commercial application	In-field efficiency (%)		
					PCS	Transformer	BESS
Parallel modular BESS	M5BAT [27,28]	6 MW/7.5MWh	5 different kinds	Frequency reserve	94.1	94.6	72.8
Cascaded modular BESS	Baoqing CHB-BESS [46]	2 MW/2MWh	Only LTO	Electric energy time-shift	98	/	90

4.2. Cost

As this review focuses on modular BESS configurations, costs due to different kinds of batteries, and protective and mechanical devices are not included in this subsection.

Wang et al. [10] carried out theoretical normalized calculations on the cost of PCSs in different modular BESS configurations with the commonly used LV 2L converter and transformer (Tx) configuration as the benchmark, and made a classified comparison as shown in Fig. 15. It shows that the transformer and large capacity filters hugely increase the total cost, especially the transformer, whose cost can account for about 40 % of the total cost. LV 2L converter+Tx, MV 2L converter, and MV 3L converter are mostly used as PCSs in parallel configurations and MDDC-BESS. The total costs of these three PCSs are similar.

For the two PCS candidates for ac-side cascaded modular BESS, CHB provides the option with the lowest cost, while MMC has the highest cost. They can save the cost of transformers and large-capacity filters. However, multi-level converters, especially MMC, need more capacitors, measurement, and communication channels due to a higher number of SMs. The required number of SMs in MMC is four times the required number for CHB in total when assuming a similar voltage rating for each SM and the converter. Fortunately, the battery pack without a parallel capacitor or with a small capacitor (several micro-farads, not expensive or bulky) can still connect to the rear-end full-bridge or half-bridge circuit to form an SM of CHB- [114,115] or MMC-BESS [40,41,106], which compensates the concern about the higher cost and larger space due to more capacitors.

4.3. Battery current ripples

Fast-switching actions will certainly induce current ripples for batteries around the switching frequency. In addition, for the SM with a single-phase ac output and without the dc-dc stage shown in Fig. 2(b), the current flowing into the battery pack also contains low-frequency ripples. For usual CHB-BESS, batteries experience both the double line frequency and switching frequency current ripples [115], while for traditional MMC-BESS, batteries even experience line frequency current ripples in addition [116]. Current ripples will influence battery current measurements and state estimations. They will also increase the additional charge throughput of batteries when they cause micro-cycles.

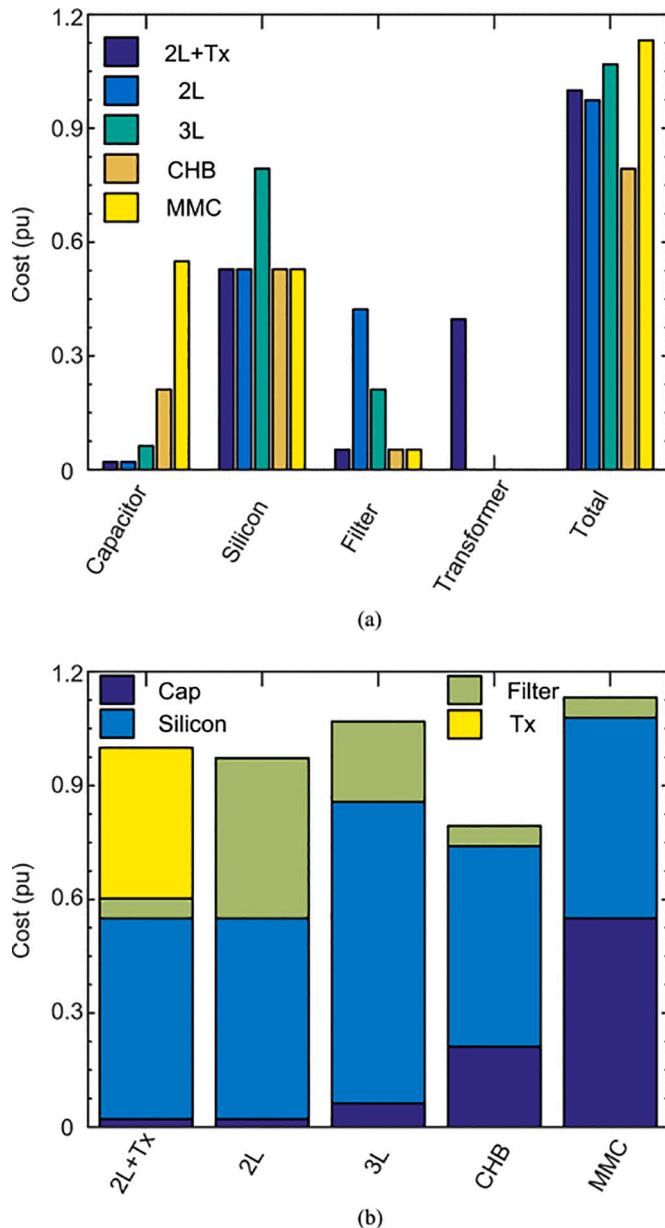


Fig. 15. Cost comparisons of different PCSs, including LV 2L converter+Tx, MV 2L converter, MV 3L converter, CHB, and MMC [10]. © [2016] IEEE. Reprinted, with permission, from Ref. [10].

However, whether they will influence battery lifetime is still an open question [117–119].

Passive/active filters are mostly used to reduce current ripples in batteries. Passive filters include single capacitor filters [36,45,63,64,70,78] and LC resonant filters [120,121]. The active filter generally refers to the dc-dc stage in each SM [37,41,106,122] to smooth the current flowing into the battery pack. However, such methods will increase system cost and size and reduce overall efficiency.

Additional control can reduce the oscillatory energy in BESS, which eliminates current ripples directly by controlling circulating currents [121,123] or injecting the common-mode voltage [124]. Shi et al. [125] moved the second-order current ripples in CHB-BESS to the fourth-order by injecting a third harmonic voltage. Ma et al. [126] also proposed a variable dc-link voltage regulation method for MMC-BESS to directly reduce the substantial additional charge throughput of batteries caused by current ripples. However, the additional control method is normally complicated and may affect the system's stability.

4.4. Electric isolation

As reviewed in Section 2, line-frequency transformers are used in parallel modular BESSs and most MDDC-BESSs. Thus, electric isolation between the MV ac grid and the battery packs can be guaranteed in these configurations.

For the transformer-less cascaded modular BESSs, the battery pack is in an MV suspension state, increasing insulation cost [3]. Fortunately, Ref. [46] mentioned that the increase in insulation cost at the medium voltage level below 10 kV is relatively limited. In addition, the high-frequency transformer used in some dc-dc stage topologies (such as DAB) can also provide electric isolation at the expense of lower efficiency, higher cost, larger weight and volume [47].

4.5. DC grid access versatility

As reviewed in Section 2, the dc-side-parallel configuration and most MDDC-BESS can connect to an LV dc distribution grid through the dc bus. MMC-BESS and the MDDC-BESS with the medium voltage dc-ac converter can access the medium voltage dc grid. The ac-side-parallel-connected modular BESS and CHB-BESS cannot be directly integrated into the dc grid.

4.6. Power flow control

As reviewed in Section 3.1, when unified dispatching is not required, each SM in the parallel modular configuration can be controlled independently with high flexibility. Also, the power flow control for this configuration is quite commercially mature.

However, once the unified power dispatching is required, the parallel modular configuration requires the power distribution strategy and an additional control center to integrate each PCS's and BMS's control system [127]. Low-latency communication is also needed to ensure control accuracy and consistency [128]. For applications such as the black start, it is also difficult to ensure that all independent SMs respond at the same time, which is prone to cause power oscillations and could even lead to black start failures [3]. Furthermore, due to the multi-layer monitoring system, complex communication architecture, and limitations on the available measurement and control devices in commercial dc-ac converters, the response speed of the parallel modular configuration can hardly reach the millisecond level required for services like providing synthetic inertia. Ref. [129] investigated the power provision time of M5BAT and found that the full feedback loop took between 450 ms and 715 ms based on the installed dc-ac converters, which is not enough to provide millisecond-level services.

In addition, the large-scale grid-tied BESS based on the parallel modular configuration has stability problems due to the possible formation of complex high-order circuits and resonance caused by control loop coupling [130,131].

The cascaded modular configuration can respond to the unified dispatching instructions as a whole with high consistency. It can also realize the complete power conversion within 2 ms and has a fast dynamic response speed [46]. However, there is a coupling relationship among SMs in the cascaded modular configuration, and the power flow must be adjusted by unified control.

4.7. Battery management

The SM number for the cascaded configuration is normally higher than the parallel configuration. When the same number of battery cells is employed, a larger number of SM results in a lower number of single cells in each SM, reducing the burden of BMS inside the SM. Furthermore, when comparing the two ac-side-cascaded modular BESS types, if assuming the same switching device type and output voltage level, MMC's SM number is four times that in CHB, which means the burden of its BMS in each SM is the lowest.

Battery balancing is the most important control for battery management. Each SM in the parallel modular configuration can independently balance battery packs, while the cascaded modular configuration needs unified hierarchical control to balance battery packs.

Comparing the reviewed battery balancing control methods in Section 3.3, MMC's unique internal circulating current provides a new method to solve the imbalances among phases and arms without injecting other external components [100], especially under grid faults [67] such as unbalanced conditions [132,133]. This method is more suitable for dealing with the balancing problems caused by the inconsistent characteristics among battery packs. However, regulating the circulating current also complicates the control of MMC-BESS.

5. Conclusions and future prospects

This paper provides a comprehensive review of four types of modular BESSs, starting with an overview of their configurations and in-field applications. It further examines and compares different control methods for these systems, including power flow control, fault-tolerant control, and battery balancing control. Notably, the paper conducts thorough performance evaluations for each configuration, yielding the following key findings:

1. BESS can significantly increase its efficiency through optimized modular design and control according to the specific application to avoid the low power/energy operation point.
2. Cascaded modular BESSs, especially CHB-BESS, demonstrate higher efficiency and lower PCS costs compared to parallel modular BESSs.
3. Parallel modular BESSs outperform cascaded modular BESSs in terms of construction and operation experience, battery current ripples, and electric isolation.
4. Regarding power flow control, parallel modular BESSs offer greater flexibility, while cascaded modular BESSs exhibit superior consistency, rapidity, and stability.
5. In battery management, parallel modular BESSs can employ independent and flexible balancing control to handle higher burdens on the BMS, whereas cascaded modular BESSs excel in unified battery balancing control. MMC-BESS stands out due to its exceptional ability to integrate and balance batteries through the use of unique circulating currents.

Based on these evaluations, the prospects for large-scale grid-tied modular BESSs encompass various aspects, including but not limited to:

1. With a significant increase in the number of BESSs connected to MV grids over the past decade, a vast amount of operating data has been collected. Evaluating the data for large-scale BESSs holds great importance, as it can provide valuable insights for future commercialization.
2. MMC-BESS still lacks solid in-field data and experience from real-world applications to verify its touted advantages, such as high efficiency, versatility in dc grid access, and strong battery integration and management capabilities.
3. Most modular BESS designs aim to reduce battery ripple currents by incorporating filters or additional control methods, leading to increased weight, cost, and control complexity. However, further research is needed to establish a clear relationship between battery aging and various characteristics of ripple currents, such as frequency and amplitude.
4. Transformer-less cascaded modular BESSs still lack a reliable method to ensure electric isolation between MV ac grids and battery packs without compromising efficiency, cost, weight, or volume.
5. Battery balancing control does not consider other battery states, such as State-of-Power, State-of-Energy, and State-of-Function. By improving the reliability of state estimation algorithms, the overall

control strategy can be optimized to maximize utilization and return on investment.

6. Despite challenges, the field of battery second-life utilization has the potential for low prices and enabling business models. Modular BESSs could help reuse second-life batteries with different SOHs in further research and engineering applications to meet the increasingly stringent requirements for utilization.

By addressing these prospects and challenges, the paper contributes to the advancement and optimization of modular BESSs, helping their development and application in the future.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT to improve the readability and language of some parts. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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References

- [1] W. Luo, S. Stynski, A. Chub, L.G. Franquelo, M. Malinowski, D. Vinnikov, Utility-scale energy storage systems: a comprehensive review of their applications, challenges, and future directions, *IEEE Ind. Electron. Mag.* 15 (4) (Dec. 2021) 17–27, <https://doi.org/10.1109/MIE.2020.3026169>.
- [2] A.A. Akhil, et al., "DOE/EPRI Electricity Storage Handbook in Collaboration with NRECA," Sandia National Lab. (SNL-NM), Albuquerque, NM (United States), SAND2015-1002, Feb. 2015, <https://doi.org/10.2172/1170618>.
- [3] M. Liu, et al., A review of power conversion systems and design schemes of high-capacity battery energy storage systems, *IEEE Access* 10 (2022) 52030–52042, <https://doi.org/10.1109/ACCESS.2022.3174193>.
- [4] T. Blank, S. Thomas, C. Roggendorf, T. Pollok, I. Trintis, D.U. Sauer, Design and construction of a test bench to characterize efficiency and reliability of high voltage battery energy storage systems, in: *Intelec 2010*, Jun. 2010, pp. 1–7, <https://doi.org/10.1109/INTELEC.2010.5525662>.
- [5] International Electrotechnical Commission, IEC 63056: Secondary Cells and Batteries Containing Alkaline or Other Non-Acid Electrolytes – Safety Requirements for Secondary Lithium Cells and Batteries for Use in Electrical Energy Storage Systems, International Electrotechnical Commission, Geneva, 2020. Accessed: Apr. 08, 2021. [Online]. Available: https://webstore.iec.ch/preview/info_iec63056%7Bed1.0%7Db.pdf.
- [6] N. Tashakor, J. Dusegimana, M. Bayati, A. Kersten, H. Schotten, S. Götz, General decoupling and sampling technique for reduced-sensor battery management systems in modular reconfigurable batteries, *Batteries* 9 (Feb. 2023) 99, <https://doi.org/10.3390/batteries9020099>.
- [7] N. Tashakor, J. Kaceti, J. Fang, Z. Li, S. Goetz, Generic dynamically reconfigurable battery with integrated auxiliary output and balancing capability, *IEEE Trans. Power Electron.* 38 (7) (Jul. 2023) 7933–7944, <https://doi.org/10.1109/TPEL.2023.3263809>.
- [8] V.F. Pires, E. Romero-Cadaval, D. Vinnikov, I. Roasto, J.F. Martins, Power converter interfaces for electrochemical energy storage systems—a review, *Energ. Convers. Manage.* 86 (2014) 453–475.

- [9] R.W.A.A. De Doncker, D.M. Divan, M.H. Kheraluwala, A three-phase soft-switched high-power-density DC/DC converter for high-power applications, *IEEE Trans. Ind. Appl.* 27 (1) (Jan. 1991) 63–73, <https://doi.org/10.1109/28.67533>.
- [10] G. Wang, et al., A review of power electronics for grid connection of utility-scale battery energy storage systems, *IEEE Trans. Sustain. Energy* 7 (4) (Oct. 2016) 1778–1790, <https://doi.org/10.1109/TSTE.2016.2586941>.
- [11] L. Chang, W. Zhang, S. Xu, K. Spence, Review on distributed energy storage systems for utility applications, *CPSS Trans. Power Electron. Appl.* 2 (4) (Dec. 2017) 267–276, <https://doi.org/10.24295/CPSSPEA.2017.00025>.
- [12] H. Hesse, M. Schimpe, D. Kucevic, A. Jossen, Lithium-ion battery storage for the grid—a review of stationary battery storage system design tailored for applications in modern power grids, *Energies* 10 (12) (2017) 2107.
- [13] L.S. Xavier, W.C.S. Amorim, A.F. Cupertino, V.F. Mendes, W.C. Do Boaventura, H. A. Pereira, Power converters for battery energy storage systems connected to medium voltage systems: a comprehensive review, *BMC Energy* 1 (1) (Dec. 2019) 7, <https://doi.org/10.1186/s42500-019-0006-5>.
- [14] S. Stynski, W. Luo, A. Chub, L.G. Franquelo, M. Malinowski, D. Vinnikov, Utility-scale energy storage systems: converters and control, *IEEE Ind. Electron. Mag.* 14 (4) (Dec. 2020) 32–52, <https://doi.org/10.1109/MIE.2020.3011655>.
- [15] F. Eroğlu, M. Kurtoğlu, A.M. Vural, Bidirectional DC–DC converter based multilevel battery storage systems for electric vehicle and large-scale grid applications: a critical review considering different topologies, state-of-charge balancing and future trends, *IET Renew. Power Gener.* 15 (5) (Apr. 2021) 915–938, <https://doi.org/10.1049/rpg2.12042>.
- [16] M. Stecca, L.R. Elizondo, T.B. Soeiro, P. Bauer, P. Palensky, A comprehensive review of the integration of battery energy storage systems into distribution networks, *IEEE Open J. Ind. Electron. Soc.* 1 (2020) 46–65, <https://doi.org/10.1109/OJIES.2020.2981832>.
- [17] M. Rouholamini, et al., A review of modeling, management, and applications of grid-connected Li-ion battery storage systems, *IEEE Trans. Smart Grid* 13 (6) (Nov. 2022) 4505–4524, <https://doi.org/10.1109/TSG.2022.3188598>.
- [18] S. Vazquez, S.M. Lukic, E. Galvan, L.G. Franquelo, J.M. Carrasco, Energy storage systems for transport and grid applications, *IEEE Trans. Ind. Electron.* 57 (12) (2010) 3881–3895.
- [19] K. Mattern, A. Ellis, S.E. Williams, C. Edwards, A. Nourai, D. Porter, Application of inverter-based systems for peak shaving and reactive power management, in: in 2008 IEEE/PES Transmission and Distribution Conference and Exposition, IEEE, Chicago, IL, USA, Apr. 2008, pp. 1–4, <https://doi.org/10.1109/TDC.2008.4517244>.
- [20] N.M.L. Tan, T. Abe, H. Akagi, Design and performance of a bidirectional isolated DC–DC converter for a battery energy storage system, *IEEE Trans. Power Electron.* 27 (3) (Mar. 2012) 1237–1248, <https://doi.org/10.1109/TPEL.2011.2108317>.
- [21] A. Hu, G. Li, T. Yang, L. Zhang, H. Shui, C. Fang, Research on the key technologies of battery energy storage power station for plug and play operation, in: 2019 IEEE Innovative Smart Grid Technologies - Asia (ISGT Asia), May 2019, pp. 2330–2335, <https://doi.org/10.1109/ISGT-Asia.2019.8881518>.
- [22] J. Muenderlein, M. Steinhoff, H. Axelsen, D.U. Sauer, Planning, building, efficiency measurement and determination of forecast data of a grid-scale hybrid 5 MW/5 MWh battery storage system, in: 2017 IEEE International Telecommunications Energy Conference (INTEC), Oct. 2017, pp. 314–320, <https://doi.org/10.1109/INTEC.2017.8214155>.
- [23] J. Muenderlein, M. Merten, T. Storek, D. Stein, I. for P. G. and S. Systems, Modularer multi-Megawatt multi-Technologie Mittelspannungs batteriespeicher (M5BAT): Abschlussbericht, in: Institute for Power Generation and Storage Systems (RWTH-Aachen University), Aachen, Germany, 2018.
- [24] K. Jacqué, L. Koltermann, J. Figgner, S. Zurmühlen, D.U. Sauer, The influence of frequency containment reserve on the operational data and the state of health of the hybrid stationary large-scale storage system, *Energies* 15 (4) (Feb. 2022) 1342, <https://doi.org/10.3390/en15041342>.
- [25] L. Koltermann, K. Jacqué, J. Figgner, S. Zurmühlen, D.U. Sauer, Operational validation of a power distribution algorithm for a modular megawatt battery storage system, *Batter. Supercaps* (2022) 1–16, <https://doi.org/10.1002/batt.202200414>.
- [26] Institute for Power Electronics and Electrical Drives (ISEA), RWTH Aachen University, EMMUseBat: Development of methods for the multi-use operation of modular large-scale battery systems in the medium-voltage grid. <https://www.isea.rwth-aachen.de/cms/ISEA/Forschung/Projekte/Oeffentliche-Projekte/Laufende-Projekte/~oyzqe/EMMUseBat/?lidx=1>. (Accessed 26 May 2023).
- [27] K. Jacqué, L. Koltermann, J. Figgner, S. Zurmühlen, D.U. Sauer, The influence of frequency containment reserve on the efficiency of a hybrid stationary large-scale storage system, *J. Energy Storage* 52 (Aug. 2022) 104961, <https://doi.org/10.1016/j.est.2022.104961>.
- [28] K. Jacqué, L. Koltermann, J. Figgner, S. Zurmühlen, D.U. Sauer, The influence of frequency containment reserve on the cycles of a hybrid stationary large-scale storage system, *J. Energy Storage* 52 (Aug. 2022) 105040, <https://doi.org/10.1016/j.est.2022.105040>.
- [29] Institute for Power Electronics and Electrical Drives (ISEA), RWTH Aachen University, Website-M5BAT. <https://m5bat.isea.rwth-aachen.de>. (Accessed 6 March 2023).
- [30] G.R. Walker, P.C. Sernia, Cascaded DC–DC converter connection of photovoltaic modules, *IEEE Trans. Power Electron.* 19 (4) (Jul. 2004) 1130–1139, <https://doi.org/10.1109/TPEL.2004.830090>.
- [31] N. Mukherjee, D. Strickland, Control of second-life hybrid battery energy storage system based on modular boost-multilevel Buck converter, *IEEE Trans. Ind. Electron.* 62 (2) (Feb. 2015) 1034–1046, <https://doi.org/10.1109/TIE.2014.2341598>.
- [32] F. Chang, Z. Zheng, Y. Li, A novel hybrid energy storage topology and its control algorithm, in: in 2014 16th European Conference on Power Electronics and Applications, Lappeenranta, Finland: IEEE, Aug. 2014, pp. 1–9, <https://doi.org/10.1109/EPE.2014.6910877>.
- [33] Z. Gao, Q. Lu, A hybrid cascaded multilevel converter based on three-level cells for battery energy management applied in electric vehicles, *IEEE Trans. Power Electron.* 34 (8) (Aug. 2019) 7326–7349, <https://doi.org/10.1109/TPEL.2018.2882851>.
- [34] L. Maharjan, S. Inoue, H. Akagi, A transformerless energy storage system based on a cascade multilevel PWM converter with star configuration, *IEEE Trans. Ind. Appl.* 44 (5) (2008) 1621–1630.
- [35] L. Maharjan, T. Yoshii, S. Inoue, H. Akagi, A transformerless energy storage system based on a cascade PWM converter with star-configuration, in: 2007 Power Conversion Conference - Nagoya, Nagoya, Japan: IEEE, 2007, pp. 751–757.
- [36] L. Maharjan, S. Inoue, H. Akagi, J. Asakura, State-of-charge (SOC) balancing control of a battery energy storage system based on a Cascaded PWM converter, *IEEE Trans. Power Electron.* 24 (6) (2009) 1628–1636.
- [37] I. Trintis, S. Munk-Nielsen, R. Teodorescu, A new modular multilevel converter with integrated energy storage, in: IECON 2011 - 37th Annual Conference of the IEEE Industrial Electronics Society, Melbourne, Vic, Australia: IEEE, Nov. 2011, pp. 1075–1080, <https://doi.org/10.1109/IECON.2011.6119457>.
- [38] I. Trintis, Grid Converters for Stationary Battery Energy Storage Systems, Department of Energy Technology, Aalborg University, 2011. Accessed: Apr. 10, 2021. [Online]. Available: <https://vbn.aau.dk/en/publications/grid-converter-s-for-stationary-battery-energy-storage-systems>.
- [39] L. Zhang, F. Gao, N. Li, Q. Zhang, C. Wang, Interlinking modular multilevel converter of hybrid AC-DC distribution system with integrated battery energy storage, in: in 2015 IEEE Energy Conversion Congress and Exposition (ECCE), Montreal, QC, Canada: IEEE, Sep. 2015, pp. 70–77, <https://doi.org/10.1109/ECCE.2015.7309671>.
- [40] L. Zhang, Y. Tang, S. Yang, F. Gao, Decoupled power control for a modular multilevel converter-based hybrid AC-DC grid integrated with hybrid energy storage, *IEEE Trans. Ind. Electron.* 66 (4) (Apr. 2019) 2926–2934, <https://doi.org/10.1109/TIE.2018.2842795>.
- [41] T. Soong, P.W. Lehn, Evaluation of emerging modular multilevel converters for BESS applications, *IEEE Trans. Power Deliv.* 29 (5) (2014) 2086–2094.
- [42] T. Soong, P.W. Lehn, Internal power flow of a modular multilevel converter with distributed energy resources, *IEEE J. Emerg. Sel. Top. Power Electron.* 2 (4) (Dec. 2014) 1127–1138, <https://doi.org/10.1109/JESTPE.2014.2342656>.
- [43] W. Jiang, C. Zhu, C. Yang, L. Zhang, S. Xue, W. Chen, The active power control of cascaded multilevel converter based hybrid energy storage system, *IEEE Trans. Power Electron.* 34 (8) (Aug. 2019) 8241–8253, <https://doi.org/10.1109/TPEL.2018.2882450>.
- [44] L. Zhang, Z. Zhang, J. Qin, D. Shi, Z. Wang, Design and performance evaluation of the modular multilevel converter (MMC)-based grid-tied PV-battery conversion system, in: In 2018 IEEE Energy Conversion Congress and Exposition (ECCE), Portland, OR, USA: IEEE, Sep. 2018, pp. 2649–2654, <https://doi.org/10.1109/ECCE.2018.8558443>.
- [45] N. Kawakami, et al., Development of a 500-kW modular multilevel cascade converter for battery energy storage systems, *IEEE Trans. Ind. Appl.* 50 (6) (Nov. 2014) 3902–3910, <https://doi.org/10.1109/TIA.2014.2313657>.
- [46] X. Cai, et al., Transformerless high-voltage power conversion system for battery energy storage system and the first demonstration application in world, *Proc. CSEE* 40 (1) (2020) 200–211.
- [47] S. Thomas, et al., HERMES: Highly Efficient and Reliable Modular Battery Energy Storage Systems, *Universitätsbibliothek der RWTH Aachen*, 2012.
- [48] M. Bragard, N. Soltan, S. Thomas, R.W.D. Doncker, The balance of renewable sources and user demands in grids: power electronics for modular battery energy storage systems, *IEEE Trans. Power Electron.* 25 (12) (Dec. 2010) 3049–3056, <https://doi.org/10.1109/TPEL.2010.2085455>.
- [49] S. Thomas, M. Stieneker, R.W.D. Doncker, Development of a modular high-power converter system for battery energy storage systems, in: Proceedings of the 2011 14th European Conference on Power Electronics and Applications, Birmingham, UK: IEEE, Aug. 2011, pp. 1–10.
- [50] A. Hillers, J. Biela, Optimal design of the modular multilevel converter for an energy storage system based on split batteries, in: in 2013 15th European Conference on Power Electronics and Applications (EPE), IEEE, Lille, France, Sep. 2013, pp. 1–11, <https://doi.org/10.1109/EPE.2013.6634660>.
- [51] A. Hillers, M. Stojadinovic, J. Biela, Systematic comparison of modular multilevel converter topologies for battery energy storage systems based on Split batteries, in: in 2015 17th European Conference on Power Electronics and Applications (EPE'15 ECCE-Europe), IEEE, Geneva, Switzerland, Sep. 2015, pp. 1–9, <https://doi.org/10.1109/EPE.2015.7309385>.
- [52] A. Hillers, H. Tu, J. Biela, Central control and distributed protection of the DSBC and DSCC modular multilevel converters, in: in 2016 IEEE Energy Conversion Congress and Exposition (ECCE), IEEE, Milwaukee, WI, USA, Sep. 2016, pp. 1–7, <https://doi.org/10.1109/ECCE.2016.7854909>.
- [53] A. Hillers, J. Biela, Increased efficiency and reduced realization effort of DSBC and DSCC modular multilevel converters (MMCs), in: in 2018 International Power Electronics Conference (IPEC-Niigata 2018 -ECCE Asia), IEEE, Niigata, Japan, May 2018, pp. 1896–1903, <https://doi.org/10.23919/IPEC.2018.8507928>.

- [54] A. Hillers, Modular Multilevel Converters (MMCs) for Grid-connected Energy Storage Systems Based on Split Batteries (sBESS), ETH Zürich, Zürich, 2018.
- [55] M.M.U. Rehman, F. Zhang, R. Zane, D. Maksimovic, Control of bidirectional DC/DC converters in reconfigurable, modular battery systems, in: 2017 IEEE Applied Power Electronics Conference and Exposition (APEC), IEEE, Tampa, FL, USA, Mar. 2017, pp. 1277–1283, <https://doi.org/10.1109/APEC.2017.7930860>.
- [56] N. Gao, R. Li, Q. Chen, X. Cai, Optimized DC-link voltage control for double-stage medium frequency isolated power conversion system, *Proc. CSEE* 35 (17) (2015) 4477–4485.
- [57] M. Choi, S. Kim, S. Seo, Energy management optimization in a battery/supercapacitor hybrid energy storage system, *IEEE Trans. Smart Grid* 3 (1) (Mar. 2012) 463–472, <https://doi.org/10.1109/TSG.2011.2164816>.
- [58] N.R. Tummuru, M.K. Mishra, S. Srinivas, Dynamic energy management of renewable grid integrated hybrid energy storage system, *IEEE Trans. Ind. Electron.* 62 (12) (Dec. 2015) 7728–7737, <https://doi.org/10.1109/TIE.2015.2455063>.
- [59] Z. Zheng, K. Wang, L. Xu, Y. Li, A hybrid cascaded multilevel converter for battery energy management applied in electric vehicles, *IEEE Trans. Power Electron.* 29 (7) (Jul. 2014) 3537–3546, <https://doi.org/10.1109/TPEL.2013.2279185>.
- [60] N. Mukherjee, D. Strickland, Control of cascaded DC–DC converter-based hybrid battery energy storage systems—part I: stability issue, *IEEE Trans. Ind. Electron.* 63 (4) (Apr. 2016) 2340–2349, <https://doi.org/10.1109/TIE.2015.2509911>.
- [61] N. Mukherjee, D. Strickland, Control of cascaded DC–DC converter-based hybrid battery energy storage systems—part II: Lyapunov approach, *IEEE Trans. Ind. Electron.* 63 (5) (May 2016) 3050–3059, <https://doi.org/10.1109/TIE.2015.2511159>.
- [62] N. Mukherjee, D. Strickland, Analysis and comparative study of different converter modes in modular second-life hybrid battery energy storage systems, *IEEE J. Emerg. Sel. Top. Power Electron.* 4 (2) (Jun. 2016) 547–563, <https://doi.org/10.1109/JESTPE.2015.2460334>.
- [63] L. Maharjan, T. Yamagishi, H. Akagi, Active-power control of individual converter cells for a battery energy storage system based on a multilevel cascade PWM converter, *IEEE Trans. Power Electron.* 27 (3) (Mar. 2012) 1099–1107, <https://doi.org/10.1109/TPEL.2010.2059045>.
- [64] C. Liu, N. Gao, X. Cai, R. Li, Differentiation power control of modules in second-life battery energy storage system based on cascaded H-bridge converter, *IEEE Trans. Power Electron.* 35 (6) (Jun. 2020) 6609–6624, <https://doi.org/10.1109/TPEL.2019.2954577>.
- [65] C. Liu, X. Cai, Q. Chen, Self-adaptation control of second-life battery energy storage system based on cascaded H-bridge converter, *IEEE J. Emerg. Sel. Top. Power Electron.* 8 (2) (Jun. 2020) 1428–1441, <https://doi.org/10.1109/JESTPE.2018.2886965>.
- [66] W. Jiang, et al., Flexible power distribution control in an asymmetrical-cascaded-multilevel-converter-based hybrid energy storage system, *IEEE Trans. Ind. Electron.* 65 (8) (Aug. 2018) 6150–6159, <https://doi.org/10.1109/TIE.2017.2787557>.
- [67] Q. Chen, R. Li, X. Cai, Analysis and fault control of hybrid modular multilevel converter with integrated battery energy storage system, *IEEE J. Emerg. Sel. Top. Power Electron.* 5 (1) (Mar. 2017) 64–78, <https://doi.org/10.1109/JESTPE.2016.2623672>.
- [68] L. Zhang, Y. Tang, S. Yang, F. Gao, A modular multilevel converter-based grid-tied battery-supercapacitor hybrid energy storage system with decoupled power control, in: 2016 IEEE 8th International Power Electronics and Motion Control Conference (IPEMC-ECCE Asia), IEEE, Hefei, China, May 2016, pp. 2964–2971, <https://doi.org/10.1109/IPEMC.2016.7512768>.
- [69] N. Li, Y. Xie, Z. Sha, F. Gao, W. Wang, Hybrid predictive control method for battery integrated modular multilevel converter, in: 2016 IEEE 8th International Power Electronics and Motion Control Conference (IPEMC-ECCE Asia), IEEE, Hefei, China, May 2016, pp. 3777–3782, <https://doi.org/10.1109/IPEMC.2016.7512899>.
- [70] G. Liang, et al., Analytical derivation of intersubmodule active power disparity limits in modular multilevel converter-based battery energy storage systems, *IEEE Trans. Power Electron.* 36 (3) (Mar. 2021) 2864–2874, <https://doi.org/10.1109/TPEL.2020.3014739>.
- [71] Z. Wang, H. Lin, Y. Ma, Improved capacitor voltage balancing control for multimode operation of modular multilevel converter with integrated battery energy storage system, *IET Power Electron.* 12 (11) (Sep. 2019) 2751–2760, <https://doi.org/10.1049/iet-pel.2019.0033>.
- [72] K. Li, L. Yuan, Z. Zhao, S. Lu, Y. Zhang, Fault-tolerant control of MMC with hot reserved submodules based on carrier phase shift modulation, *IEEE Trans. Power Electron.* 32 (9) (Sep. 2017) 6778–6791, <https://doi.org/10.1109/TPEL.2016.2628762>.
- [73] B. Li, Y. Zhang, R. Yang, R. Xu, D. Xu, W. Wang, Seamless transition control for modular multilevel converters when inserting a cold-reserve redundant submodule, *IEEE Trans. Power Electron.* 30 (8) (Aug. 2015) 4052–4057, <https://doi.org/10.1109/TPEL.2015.2392778>.
- [74] K. Bi, Q. An, J. Duan, L. Sun, K. Gai, Fast diagnostic method of open circuit fault for modular multilevel DC/DC converter applied in energy storage system, *IEEE Trans. Power Electron.* 32 (5) (May 2017) 3292–3296, <https://doi.org/10.1109/TPEL.2016.2646402>.
- [75] G. Liu, Z. Xu, Y. Xue, G. Tang, Optimized control strategy based on dynamic redundancy for the modular multilevel converter, *IEEE Trans. Power Electron.* 30 (1) (Jan. 2015) 339–348, <https://doi.org/10.1109/TPEL.2014.2305663>.
- [76] W. Song, A.Q. Huang, Fault-tolerant design and control strategy for cascaded H-bridge multilevel converter-based STATCOM, *IEEE Trans. Ind. Electron.* 57 (8) (Aug. 2010) 2700–2708, <https://doi.org/10.1109/TIE.2009.2036019>.
- [77] P. Hu, D. Jiang, Y. Zhou, Y. Liang, J. Guo, Z. Lin, Energy-balancing control strategy for modular multilevel converters under submodule fault conditions, *IEEE Trans. Power Electron.* 29 (9) (Sep. 2014) 5021–5030, <https://doi.org/10.1109/TPEL.2013.2284919>.
- [78] L. Maharjan, T. Yamagishi, H. Akagi, J. Asakura, Fault-tolerant operation of a battery-energy-storage system based on a multilevel cascade PWM converter with star configuration, *IEEE Trans. Power Electron.* 25 (9) (Sep. 2010) 2386–2396, <https://doi.org/10.1109/TPEL.2010.2047407>.
- [79] F. Carnielutti, H. Pinheiro, C. Rech, Generalized carrier-based modulation strategy for cascaded multilevel converters operating under fault conditions, *IEEE Trans. Ind. Electron.* 59 (2) (Feb. 2012) 679–689, <https://doi.org/10.1109/TIE.2011.2157289>.
- [80] J.I.Y. Ota, T. Sato, H. Akagi, Enhancement of performance, availability, and flexibility of a battery energy storage system based on a modular multilevel cascaded converter (MMCC-SSBC), *IEEE Trans. Power Electron.* 31 (4) (Apr. 2016) 2791–2799, <https://doi.org/10.1109/TPEL.2015.2450757>.
- [81] M. Aleenejad, H. Mahmoudi, R. Ahmadi, A fault-tolerant strategy based on fundamental phase-shift compensation for three-phase multilevel converters with quasi-Z-source networks with discontinuous input current, *IEEE Trans. Power Electron.* 31 (11) (Nov. 2016) 7480–7488, <https://doi.org/10.1109/TPEL.2016.2520884>.
- [82] Mohammad Ali Paymani, Mohammad Saleh Marhaba, H. Iman-Eini, Fault-tolerant operation of a medium voltage drive based on the cascaded H-bridge inverter, in: 2011 2nd Power Electronics, Drive Systems and Technologies Conference, IEEE, Tehran, Iran, Feb. 2011, pp. 551–556, <https://doi.org/10.1109/PEDSTC.2011.5742480>.
- [83] A. Hillers, J. Biela, Fault-tolerant operation of the modular multilevel converter in an energy storage system based on Split batteries, in: 2014 16th European Conference on Power Electronics and Applications, IEEE, Lappeenranta, Finland, Aug. 2014, pp. 1–8, <https://doi.org/10.1109/EPE.2014.6910952>.
- [84] F. Deng, Y. Tian, R. Zhu, Z. Chen, Fault-tolerant approach for modular multilevel converters under submodule faults, *IEEE Trans. Ind. Electron.* 63 (11) (Nov. 2016) 7253–7263, <https://doi.org/10.1109/TIE.2016.2538201>.
- [85] K. Li, Z. Zhao, L. Yuan, S. Lu, B. Pan, Z. Lu, Fault tolerant control of MMC with redundant submodules based on carrier phase shift modulation, in: 2016 IEEE Applied Power Electronics Conference and Exposition (APEC), IEEE, Long Beach, CA, USA, Mar. 2016, pp. 2613–2619, <https://doi.org/10.1109/APEC.2016.7468233>.
- [86] J. Wang, Y. Tang, A fault-tolerant operation method for medium voltage modular multilevel converters with phase-shifted carrier modulation, *IEEE Trans. Power Electron.* 34 (10) (Oct. 2019) 9459–9470, <https://doi.org/10.1109/TPEL.2018.2890536>.
- [87] F. Gao, X. Gu, Z. Ma, C. Zhang, Redistributed Pulsewidth modulation of MMC battery energy storage system under submodule fault condition, *IEEE Trans. Power Electron.* 35 (3) (Mar. 2020) 2284–2294, <https://doi.org/10.1109/TPEL.2019.2925284>.
- [88] X. Gu, F. Gao, F. Aamir, X. Liu, J. Xiao, Redistributed pulse width modulation of MMC battery energy storage system under submodule fault condition, in: 2017 IEEE Energy Conversion Congress and Exposition (ECCE), IEEE, Cincinnati, OH, Oct. 2017, pp. 4171–4176, <https://doi.org/10.1109/ECCE.2017.8096723>.
- [89] M. Lelie, et al., Battery management system hardware concepts: an overview, *Appl. Sci.* 8 (4) (Mar. 2018) 534, <https://doi.org/10.3390/app8040534>.
- [90] M. Angulo, P. Lezana, S. Kouro, J. Rodriguez, B. Wu, Level-shifted PWM for cascaded multilevel inverters with even power distribution, in: 2007 IEEE Power Electronics Specialists Conference, IEEE, Orlando, FL, USA, Jun. 2007, pp. 2373–2378, <https://doi.org/10.1109/PESC.2007.4342382>.
- [91] K. Shen, et al., A modulation reconfiguration based fault-tolerant control scheme for modular multilevel converters, in: 2013 Twenty-Eighth Annual IEEE Applied Power Electronics Conference and Exposition (APEC), IEEE, Long Beach, CA, USA, Mar. 2013, pp. 3251–3255, <https://doi.org/10.1109/APEC.2013.6520766>.
- [92] C. Young, N. Chu, L. Chen, Y. Hsiao, C. Li, A single-phase multilevel inverter with battery balancing, *IEEE Trans. Ind. Electron.* 60 (5) (May 2013) 1972–1978, <https://doi.org/10.1109/TIE.2012.2207656>.
- [93] B. Ge, Y. Liu, H. Abu-Rub, F.Z. Peng, State-of-charge balancing control for a battery-energy-stored quasi-Z-source cascaded-multilevel-inverter-based photovoltaic power system, *IEEE Trans. Ind. Electron.* 65 (3) (Mar. 2018) 2268–2279, <https://doi.org/10.1109/TIE.2017.2745406>.
- [94] Z. Ma, X. Yi, W. Li, F. Gao, F. Yu, Sorting selection balancing control for the modular multilevel DC/DC converter in battery swapping stations, in: 2021 IEEE Energy Conversion Congress and Exposition (ECCE), Vancouver, BC, Canada, Oct. 2021, pp. 1730–1734, <https://doi.org/10.1109/ECCE47101.2021.9595936>.
- [95] H. Iman-Eini, J. Schanen, S. Farhangi, J. Roudet, A modular strategy for control and voltage balancing of cascaded H-bridge rectifiers, *IEEE Trans. Power Electron.* 23 (5) (Sep. 2008) 2428–2442, <https://doi.org/10.1109/TPEL.2008.2002055>.
- [96] E. Chatzinikolaou, D.J. Rogers, Cell SoC balancing using a cascaded full-bridge multilevel converter in battery energy storage systems, *IEEE Trans. Ind. Electron.* 63 (9) (Sep. 2016) 5394–5402, <https://doi.org/10.1109/TIE.2016.2565463>.
- [97] W. Jiang, L. Huang, L. Zhang, H. Zhao, L. Wang, W. Chen, Control of active power exchange with auxiliary power loop in a single-phase cascaded multilevel converter-based energy storage system, *IEEE Trans. Power Electron.* 32 (2) (Feb. 2017) 1518–1532, <https://doi.org/10.1109/TPEL.2016.2543751>.
- [98] G. Liang, et al., A constrained Intersubmodule state-of-charge balancing method for battery energy storage systems based on the cascaded H-bridge converter,

- IEEE Trans. Power Electron. 37 (10) (2022) 12669–12678, <https://doi.org/10.1109/TPEL.2022.3170062>.
- [99] J.A. Barrera, L. Marrojo, M.A.R. Vidal, J.R.T. Apraiz, Individual voltage balancing strategy for PWM cascaded H-bridge converter-based STATCOM, IEEE Trans. Ind. Electron. 55 (1) (Jan. 2008) 21–29, <https://doi.org/10.1109/TIE.2007.906127>.
- [100] F. Gao, L. Zhang, Q. Zhou, M. Chen, T. Xu, S. Hu, State-of-charge balancing control strategy of battery energy storage system based on modular multilevel converter, in: in 2014 IEEE Energy Conversion Congress and Exposition (ECCE), IEEE, Pittsburgh, PA, USA, Sep. 2014, pp. 2567–2574, <https://doi.org/10.1109/ECCE.2014.6953744>.
- [101] K. Bi, L. Sun, Q. An, J. Duan, Active SOC balancing control strategy for modular multilevel super capacitor energy storage system, IEEE Trans. Power Electron. 34 (5) (May 2019) 4981–4992, <https://doi.org/10.1109/TPEL.2018.2865553>.
- [102] S. Chowdhury, M.N. Bin Shaheed, Y. Sozer, An integrated state of health (SOH) balancing method for lithium-ion battery cells, in: 2019 IEEE Energy Conversion Congress and Exposition (ECCE), Sep. 2019, pp. 5759–5763, <https://doi.org/10.1109/ECCE.2019.8912932>.
- [103] N. Mukherjee, D. De, A new state-of-charge control derivation method for hybrid battery type integration, IEEE Trans. Energy Convers. 32 (3) (Sep. 2017) 866–875, <https://doi.org/10.1109/TEC.2017.2695242>.
- [104] M. Vasiladiotis, A. Rufer, Balancing control actions for cascaded H-bridge converters with integrated battery energy storage, in: 2013 15th European Conference on Power Electronics and Applications (EPE), Sep. 2013, pp. 1–10, <https://doi.org/10.1109/EPE.2013.6634337>.
- [105] A. Lachichi, Modular multilevel converters with integrated batteries energy storage, in: in 2014 International Conference on Renewable Energy Research and Application (ICRERA), IEEE, Milwaukee, WI, USA, Oct. 2014, pp. 828–832, <https://doi.org/10.1109/ICRERA.2014.7016501>.
- [106] M. Vasiladiotis, A. Rufer, Analysis and control of modular multilevel converters with integrated battery energy storage, IEEE Trans. Power Electron. 30 (1) (Jan. 2015) 163–175, <https://doi.org/10.1109/TPEL.2014.2303297>.
- [107] N. Hatano, T. Ise, A configuration and control method of cascade H-bridge STATCOM, in: 2008 IEEE Power and Energy Society General Meeting - Conversion and Delivery of Electrical Energy in the 21st Century, Jul. 2008, pp. 1–8, <https://doi.org/10.1109/PES.2008.4596240>.
- [108] N. Hatano, T. Ise, Control scheme of cascaded H-bridge STATCOM using zero-sequence voltage and negative-sequence current, IEEE Trans. Power Deliv. 25 (2) (Apr. 2010) 543–550, <https://doi.org/10.1109/TPWRD.2009.2035221>.
- [109] N. Li, F. Gao, Linearized operation of MMC battery energy storage system, in: in 2016 IEEE 2nd Annual Southern Power Electronics Conference (SPEC), IEEE, Auckland, New Zealand, Dec. 2016, pp. 1–5, <https://doi.org/10.1109/SPEC.2016.7846013>.
- [110] N. Li, F. Gao, T. Hao, Z. Ma, C. Zhang, SOH balancing control method for the MMC battery energy storage system, IEEE Trans. Ind. Electron. 65 (8) (Aug. 2018) 6581–6591, <https://doi.org/10.1109/TIE.2017.2733462>.
- [111] Z. Ma, F. Yu, X. Zhao, F. Gao, The multidimensional battery management strategy for MMC battery energy storage system, in: 2022 IEEE 7th Southern Power Electronics Conference (SPEC), Dec. 2022, pp. 1–5, <https://doi.org/10.1109/SPEC55080.2022.10058251>.
- [112] M. Chen, B. Zhang, Y. Li, G. Qi, J. Liu, Design of a Multi-Level Battery Management System for a Cascade H-bridge energy storage system, in: in 2014 IEEE PES Asia-Pacific Power and Energy Engineering Conference (APPEEC), Hong Kong, IEEE, China, Dec. 2014, pp. 1–5, <https://doi.org/10.1109/APPEEC.2014.7066076>.
- [113] Z. Ma, et al., Multilayer SOH equalization scheme for MMC battery energy storage system, IEEE Trans. Power Electron. 35 (12) (Dec. 2020) 13514–13527, <https://doi.org/10.1109/TPEL.2020.2991879>.
- [114] E. Chatzinikolaou, D.J. Rogers, Hierarchical distributed balancing control for large-scale reconfigurable AC battery packs, IEEE Trans. Power Electron. 33 (7) (Jul. 2018) 5592–5602, <https://doi.org/10.1109/TPEL.2017.2782675>.
- [115] F. Chang, F. Roemer, M. Lienkamp, Influence of current ripples in cascaded multilevel topologies on the aging of lithium batteries, IEEE Trans. Power Electron. 35 (11) (Nov. 2020) 11879–11890, <https://doi.org/10.1109/TPEL.2020.2989145>.
- [116] I. Puranik, L. Zhang, J. Qin, Impact of low-frequency ripple on lifetime of battery in MMC-based battery storage systems, in: in 2018 IEEE Energy Conversion Congress and Exposition (ECCE), IEEE, Portland, OR, USA, Sep. 2018, pp. 2748–2752, <https://doi.org/10.1109/ECCE.2018.8558061>.
- [117] L.W. Juang, P.J. Kollmeyer, A.E. Anders, T.M. Jahns, R.D. Lorenz, D. Gao, Investigation of the influence of superimposed AC current on lithium-ion battery aging using statistical design of experiments, J. Energy Storage 11 (Jun. 2017) 93–103, <https://doi.org/10.1016/j.est.2017.02.005>.
- [118] A. Bessman, R. Soares, O. Wallmark, P. Svens, G. Lindbergh, Aging effects of AC harmonics on Lithium-ion cells, J. Energy Storage 21 (Feb. 2019) 741–749, <https://doi.org/10.1016/j.est.2018.12.016>.
- [119] E. Goldammer, et al., The impact of an overlaid ripple current on battery aging: the development of the SiCWell dataset, Batteries 8 (2) (Jan. 2022) 11, <https://doi.org/10.3390/batteries8020011>.
- [120] M. Vasiladiotis, Modular Multilevel Converters with Integrated Split Battery Energy Storage, EPFL, Lausanne, 2014.
- [121] S.B. Wersland, A.B. Acharya, L.E. Norum, Integrating battery into MMC submodule using passive technique, in: in 2017 IEEE 18th Workshop on Control and Modeling for Power Electronics (COMPEL), IEEE, Stanford, CA, USA, Jul. 2017, pp. 1–7, <https://doi.org/10.1109/COMPEL.2017.8013309>.
- [122] I. Trintis, S. Munk-Nielsen, R. Teodorescu, Cascaded H-bridge with bidirectional boost converters for energy storage, in: Proceedings of the 2011 14th European Conference on Power Electronics and Applications, IEEE, Birmingham, UK, 2011, pp. 1–9.
- [123] Z. Li, R. Lizana, S.M. Lukic, A.V. Peterchev, S.M. Goetz, Current injection methods for ripple-current suppression in delta-configured Split-battery energy storage, IEEE Trans. Power Electron. 34 (8) (Aug. 2019) 7411–7421, <https://doi.org/10.1109/TPEL.2018.2879613>.
- [124] Z. Li, R. Lizana, A.V. Peterchev, S.M. Goetz, Ripple current suppression methods for star-configured modular multilevel converters, in: IECON 2017-43rd Annual Conference of the IEEE Industrial Electronics Society, IEEE, Beijing, China, Oct. 2017, pp. 1505–1510, <https://doi.org/10.1109/IECON.2017.8216256>.
- [125] X. Shi, C. Zhang, R. Li, X. Cai, Control scheme for second harmonic current elimination in single-star configuration based cascade multilevel energy storage system, IEEE J. Emerg. Sel. Top. Ind. Electron. 3 (4) (Oct. 2022) 1210–1216, <https://doi.org/10.1109/JESTIE.2022.3195471>.
- [126] Z. Ma, F. Gao, C. Zhang, W. Li, D. Niu, Variable DC-link voltage regulation of single-phase MMC battery energy-storage system for reducing additional charge throughput, IEEE Trans. Power Electron. 36 (12) (Dec. 2021) 14267–14281, <https://doi.org/10.1109/TPEL.2021.3084605>.
- [127] M.T. Lawder, et al., Battery energy storage system (BESS) and battery management system (BMS) for grid-scale applications, Proc. IEEE 102 (6) (Jun. 2014) 1014–1030, <https://doi.org/10.1109/JPROC.2014.2317451>.
- [128] J. He, Y.W. Li, F. Blaabjerg, An enhanced islanding microgrid reactive power, imbalance power, and harmonic power sharing scheme, IEEE Trans. Power Electron. 30 (6) (Jun. 2015) 3389–3401, <https://doi.org/10.1109/TPEL.2014.2332998>.
- [129] L. Koltermann, et al., Potential analysis of current battery storage systems for providing fast grid services like synthetic inertia – case study on a 6 MW system, J. Energy Storage 57 (Jan. 2023) 106190, <https://doi.org/10.1016/j.est.2022.106190>.
- [130] D. Dong, B. Wen, P. Mattavelli, D. Boroyevich, Y. Xue, Grid-synchronization modeling and its stability analysis for multi-parallel three-phase inverter systems, in: in 2013 Twenty-Eighth Annual IEEE Applied Power Electronics Conference and Exposition (APEC), IEEE, Long Beach, CA, USA, Mar. 2013, pp. 439–446, <https://doi.org/10.1109/APEC.2013.6520247>.
- [131] R. Beres, X. Wang, F. Blaabjerg, C.L. Bak, M. Liserre, Comparative evaluation of passive damping topologies for parallel grid-connected converters with LCL filters, in: in 2014 International Power Electronics Conference (IPEC-Hiroshima 2014 - ECCE ASIA), IEEE, Hiroshima, Japan, May 2014, pp. 3320–3327, <https://doi.org/10.1109/IPEC.2014.6870163>.
- [132] M. Vasiladiotis, N. Cherix, A. Rufer, Impact of grid asymmetries on the operation and capacitive energy storage design of modular multilevel converters, IEEE Trans. Ind. Electron. 62 (11) (2015) 6697–6707.
- [133] L. Zhang, F. Gao, N. Li, Control strategy of MMC battery energy storage system under asymmetrical grid voltage condition, Chin. J. Electr. Eng. 2 (2) (2017) 76–83.